CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

Generated on 2026-10-16 14:51

Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	100
Stimuli	2
Observed cells	5413
Timepoints	0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432
Cell types	Neural, Stromal, Trophoblast, iPSC, Epithelial, MEF
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (observed / depth factor, model draws at the reference depth)

Perturbations (perturbation_simulation)

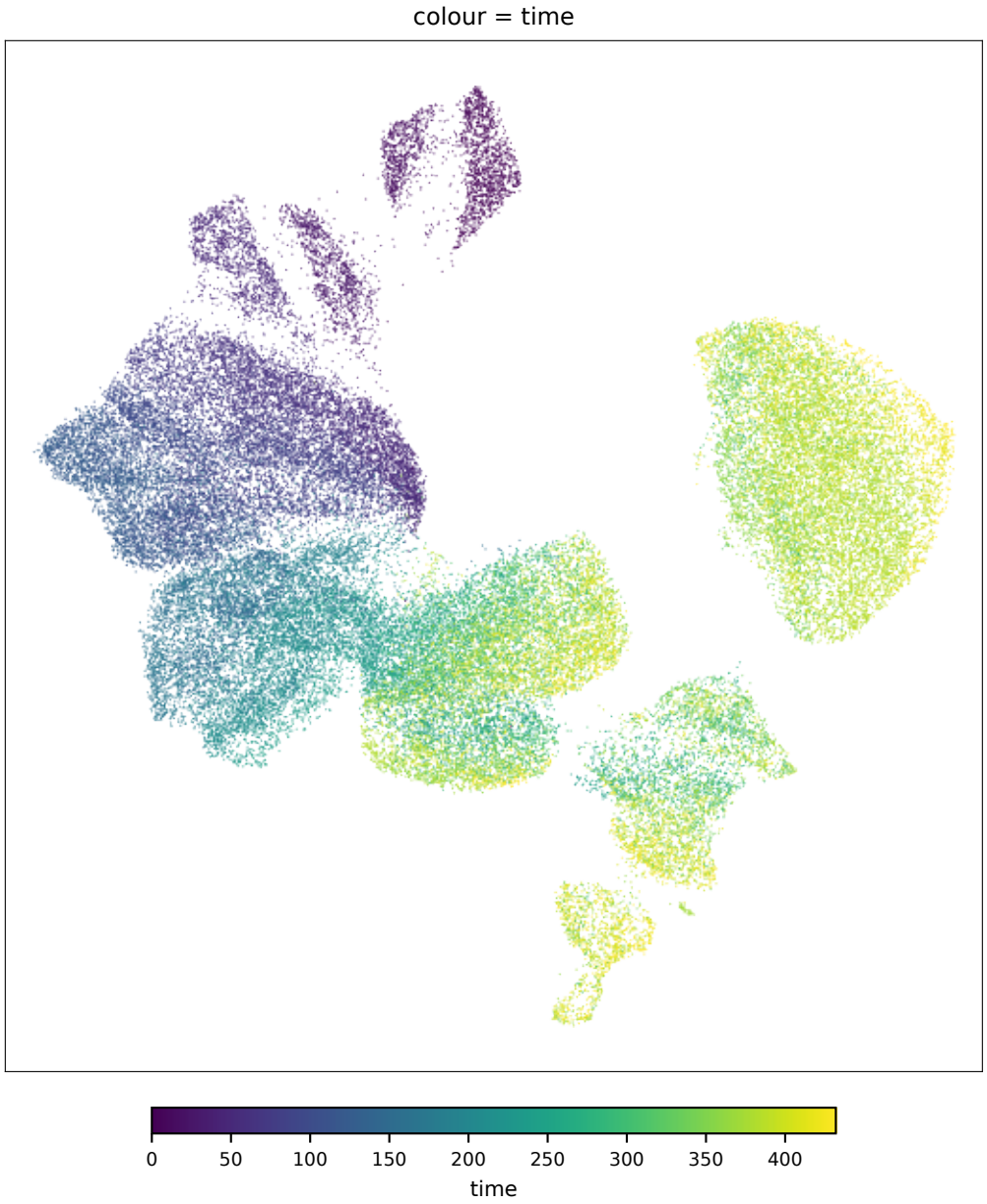
- ✓ P1 OV Obox6
- ✓ P2 OV Zfp42

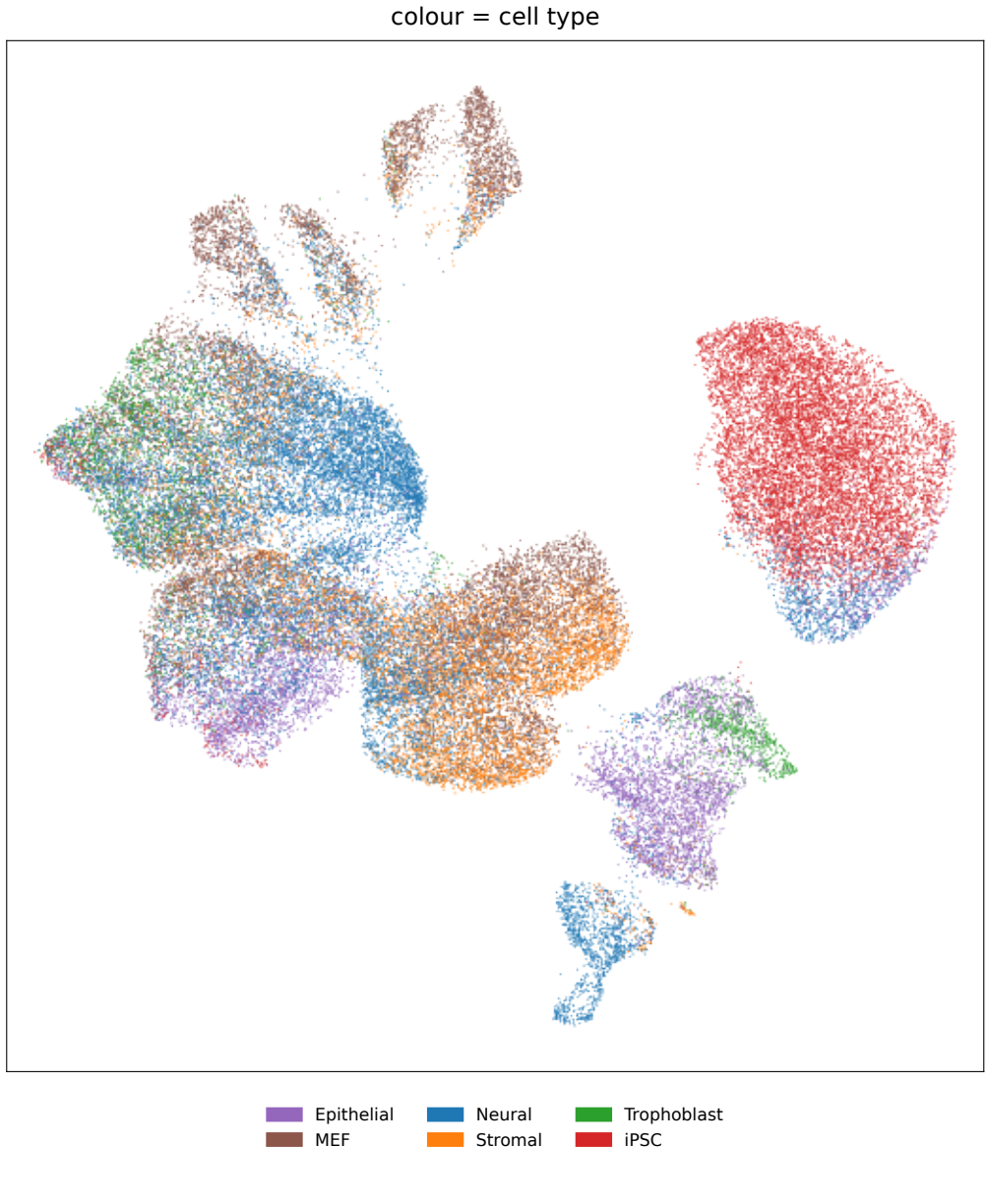
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples); teaser: classical OT vs CardamomOT displacement fields (all times, then per interval), then cell-type transitio
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2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

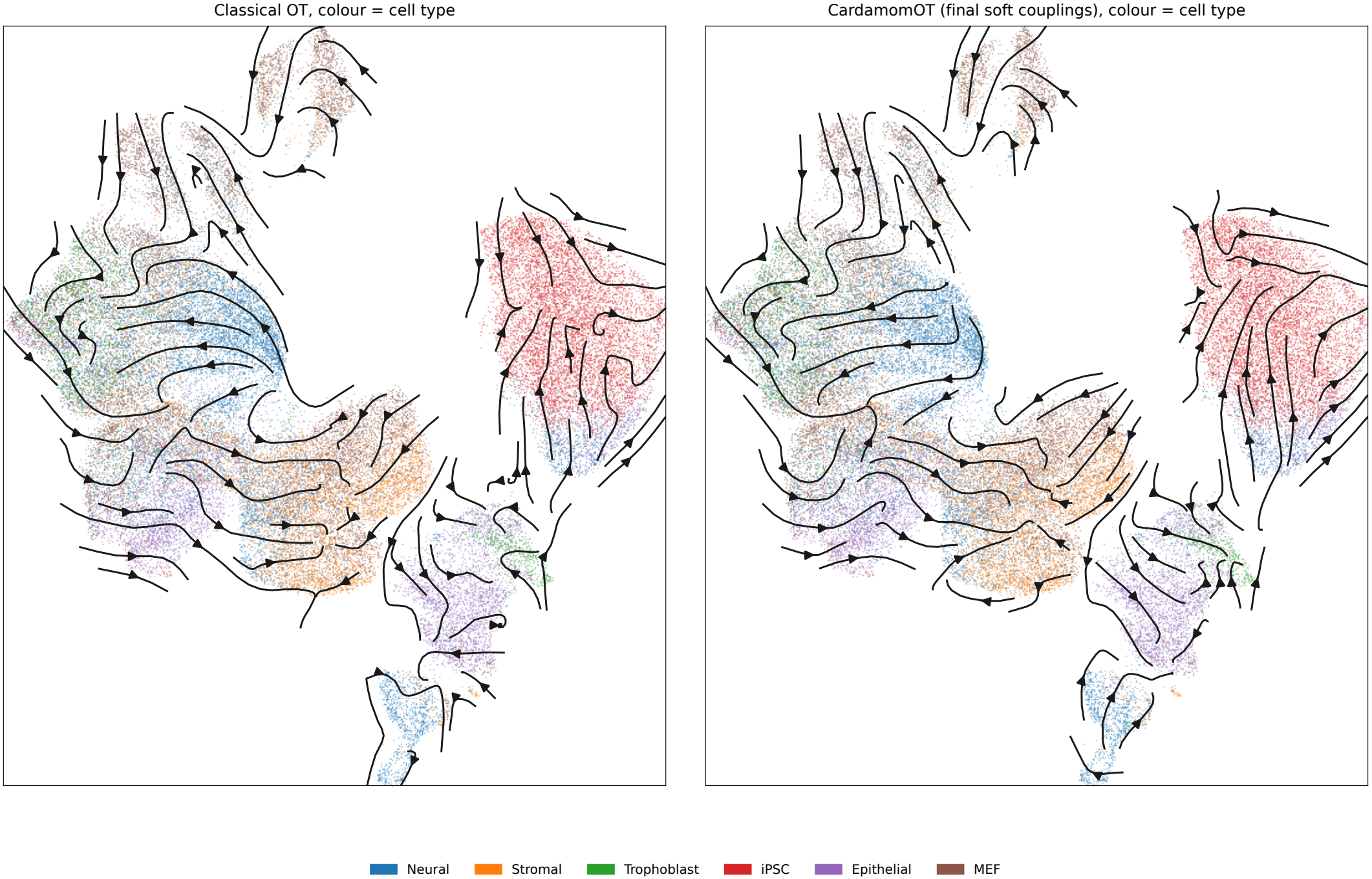
Data/data.h5ad, 50897 cells: log1p(counts), 2000 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





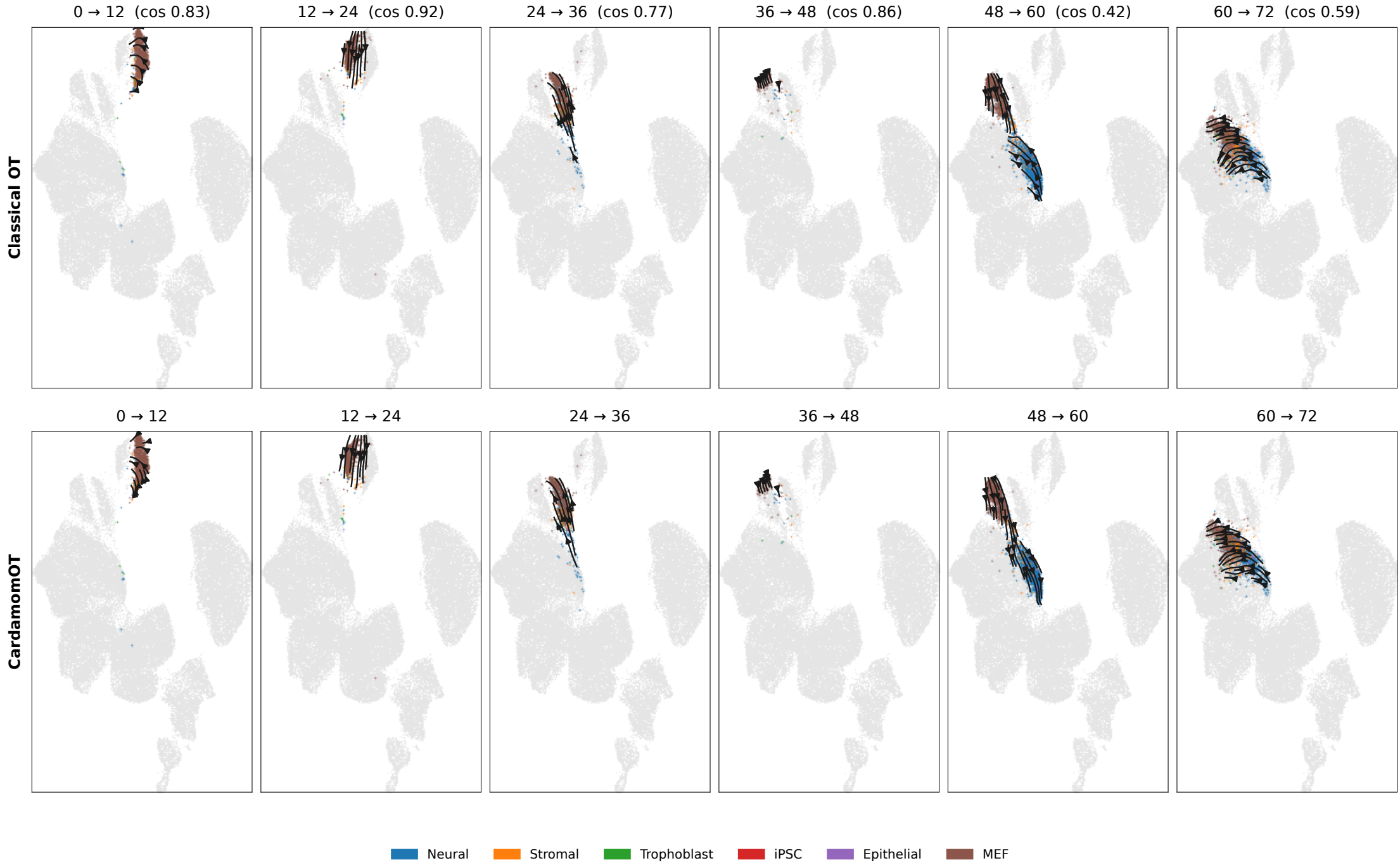
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.62 (PCA space). CardamomOT: final soft couplings of the trajectories (5413 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



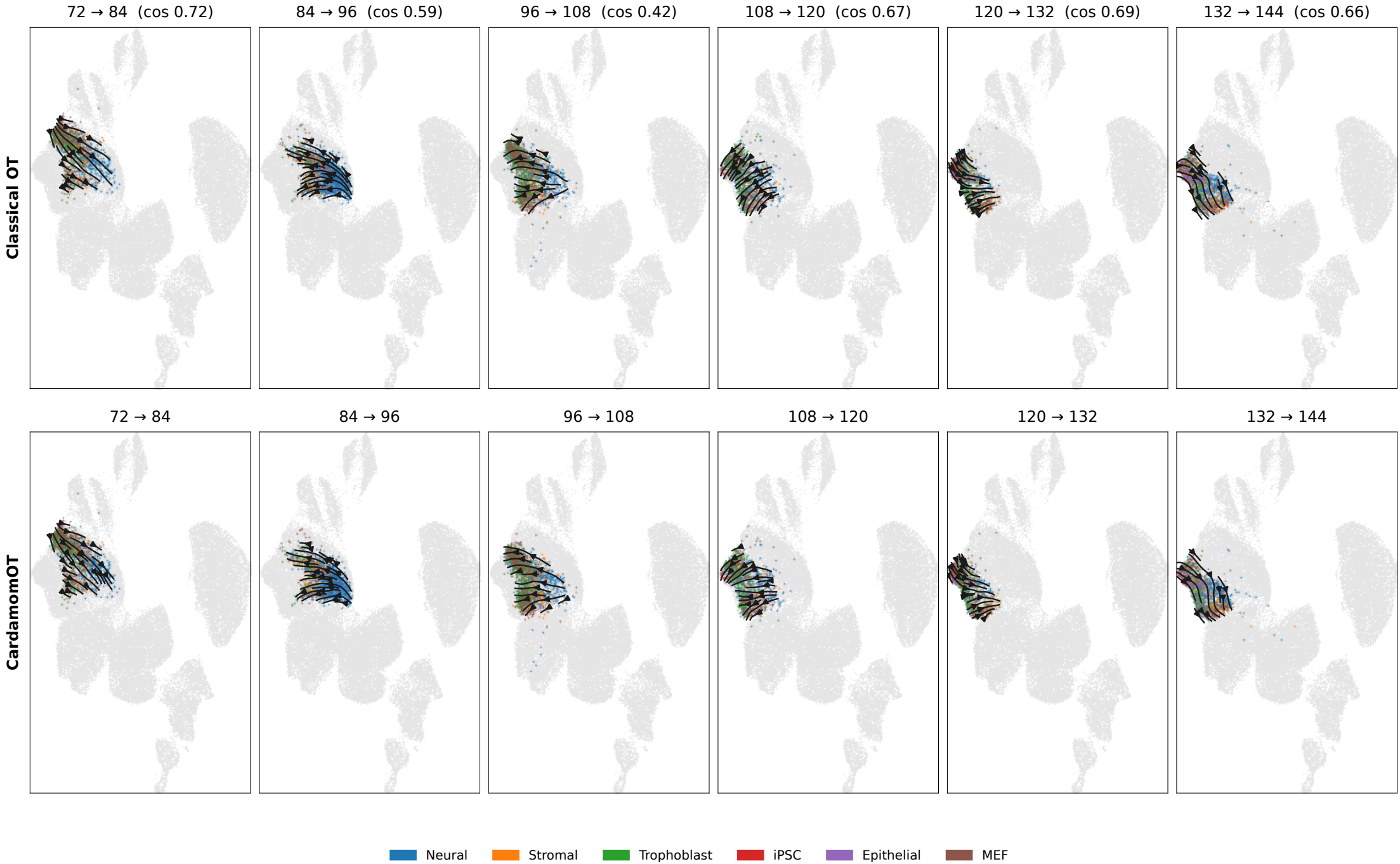
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



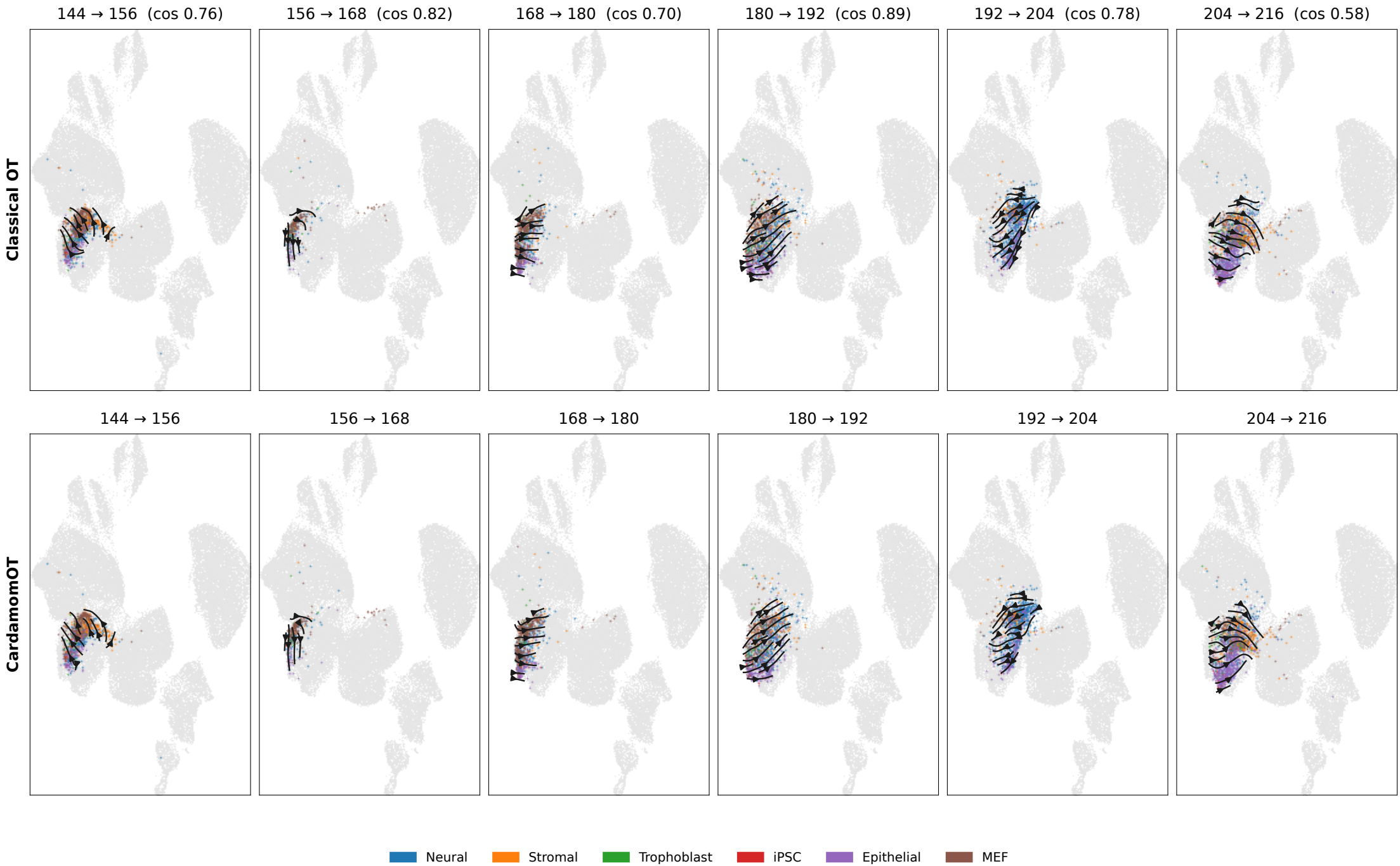
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



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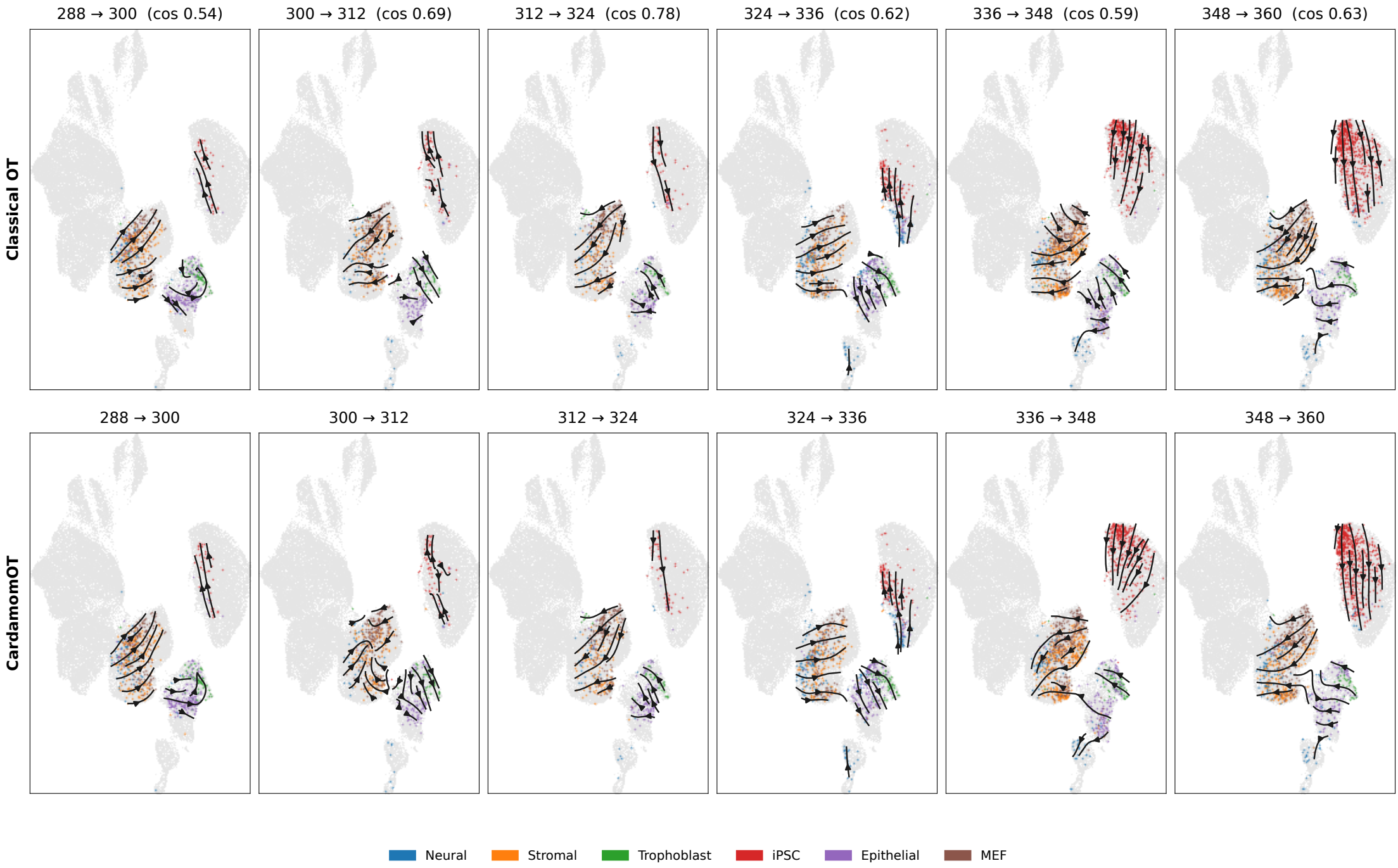
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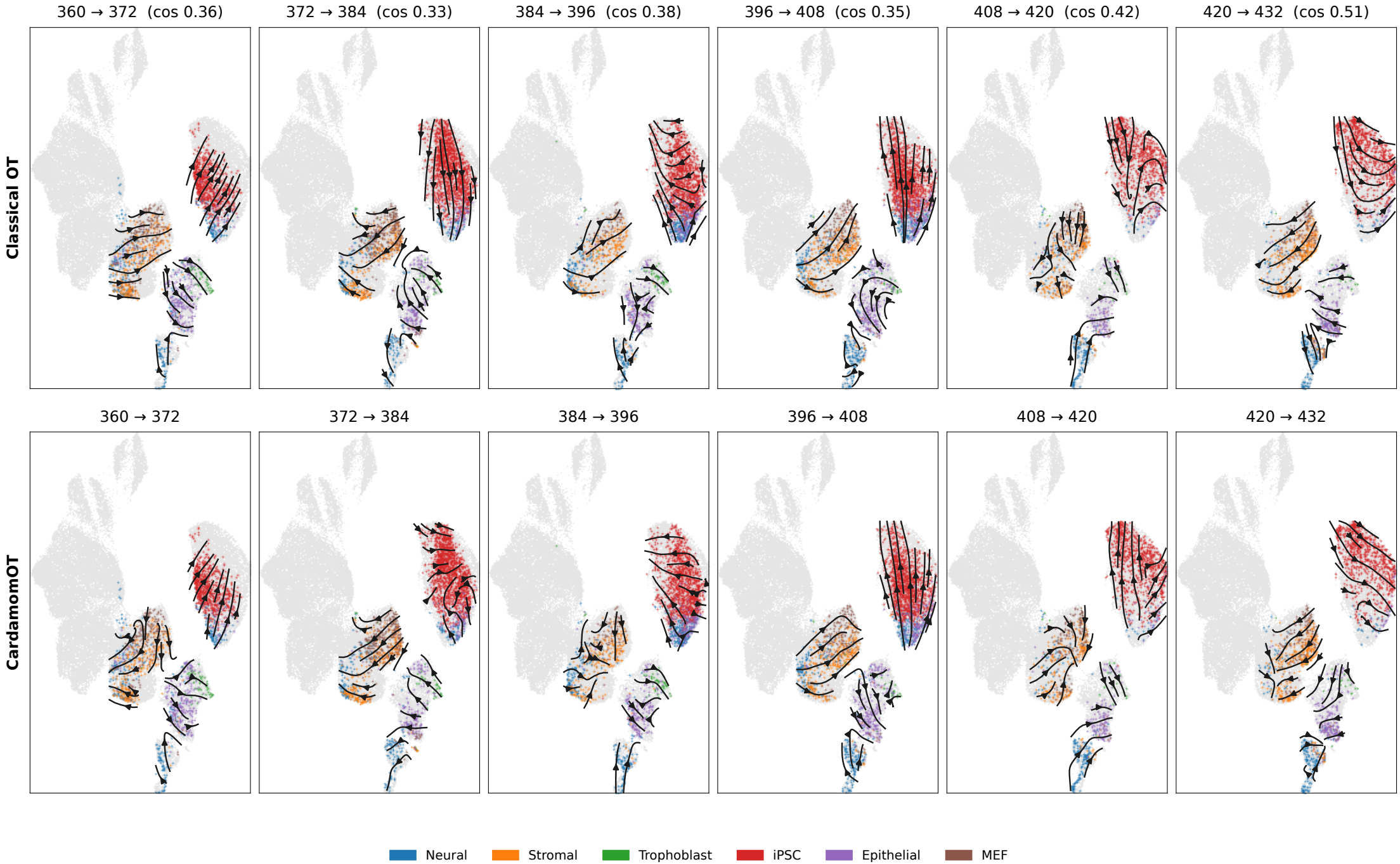
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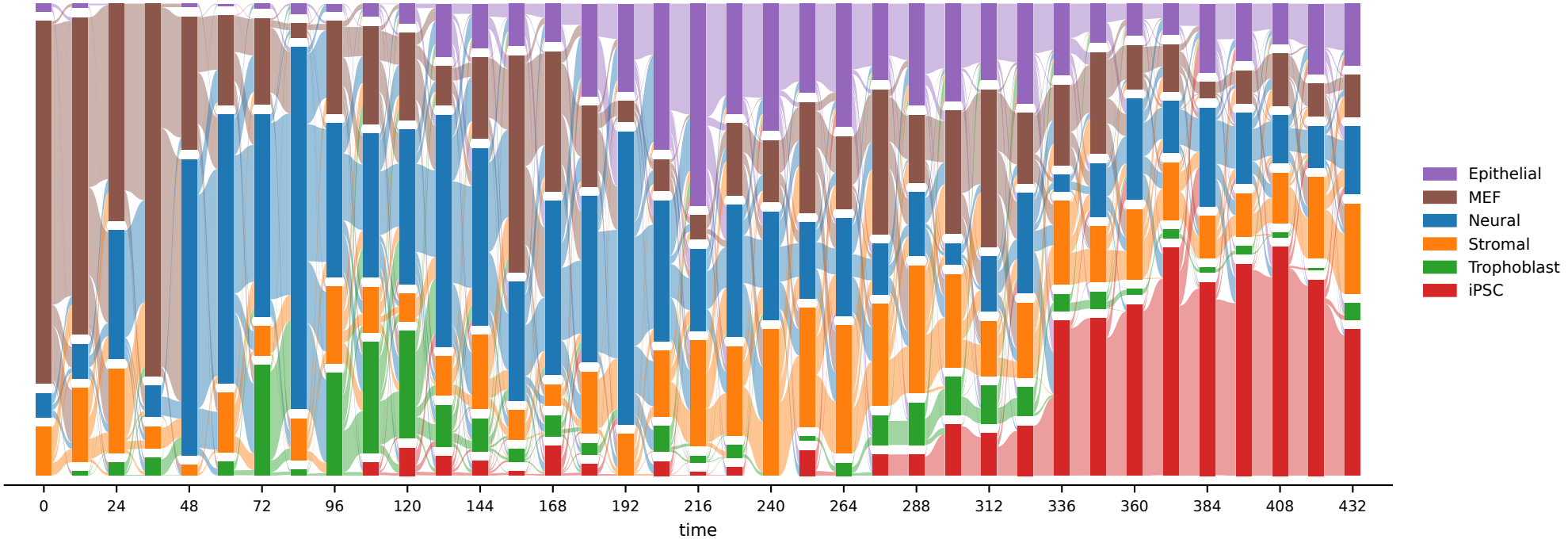
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

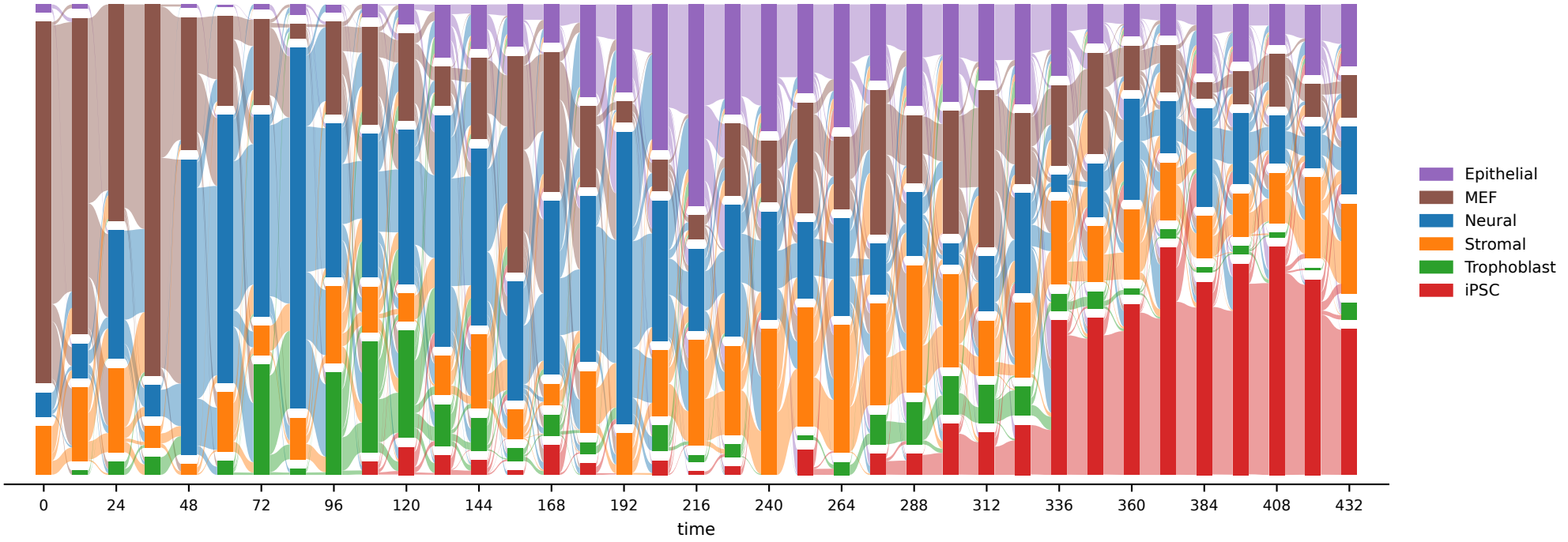
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Bex1** **Krt18** Bex4 **Spint2** **Cldn6** **Epcam** **Cldn7** **Krt19** **Cdh1**	**Krt8** **Krt18** **Epcam** **Cldn7** **Krt19** **Cldn6** **Cdh1** **Spint2** **Bex1** Dsp
MEF	**Col1a2** **Cald1** Timp1 **Thbs1** **Tagln** Col1a1 Lox Vim Timp2 **S100a6**	**Cald1** **S100a6** **Col1a2** Lmna Fstl1 Ahnak Flna Itgb1 **Tagln** **Thbs1**
Neural	**Vim** Cd109 **Cd24a** **Ccnd2** **S100a6** Limd2 **Avpi1** **S100a10** **Hmgn3** **Id1**	**Vim** **Ccnd2** **Id1** **Cd24a** H2afz **S100a6** **Avpi1** Tmsb4x **S100a10** **Hmgn3**
Stromal	**Sparc** **Lgals1** **Mgp** **Vim** **Bgn** **Col1a2** Pcolce **Col1a1** Aebp1 **Serpinh1**	**Sparc** **Lgals1** **Vim** **Mgp** **Serpinh1** **Col1a2** **Bgn** Gabarap Emp3 **Col1a1**
Trophoblast	**Ranbp1** **Mrpl12** **Cycs** **Tagln2** H2afz **Lgals3** Hsp90aa1 **Sri** Cks1b Nhp2	Dbi **Cycs** **Sri** Gng12 Anxa2 **Lgals3** **Tagln2** **Ranbp1** **Mrpl12** S100a6
iPSC	**Dppa5a** **Hsp90aa1** **Chchd10** **Mkrn1** **Apoe** **Bex1** **Fkbp3** **Mt2** **Fabp3** **Slc2a3**	**Dppa5a** **Hsp90aa1** **Chchd10** **Mkrn1** **Bex1** **Apoe** **Fkbp3** **Mt2** **Fabp3** **Slc2a3**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Epcam** **Krt18** **Krt7** **Mbnl3** **S100a14** **Bbx** Dbi Hmgb2 **Marcksl1**	**Krt8** **Epcam** **Krt18** **Krt7** **Mbnl3** **S100a14** **Bbx** **Marcksl1** Cdkn1c Hcfc1r1
MEF	**Col1a2** **Thbs1** **Vim** **Emp3** Sparc **Tagln** **Tpm1** Col1a1 **Flna** Timp2	**Vim** **Tpm1** **Flna** **Col1a2** **Emp3** Cd44 **Tagln** **Thbs1** Ccnd2 Xist
Neural	**Vim** **Pcolce2** **Ccnd2** **Myc** Ybx3 **Cenpa** **Dmrtc2** S100a10 **Cks1b** Ddx21	**Dmrtc2** **Vim** **Ccnd2** Id3 **Myc** **Pcolce2** **Cenpa** Xist **Cks1b** Rrm1
Stromal	**Mgp** **Sparc** **Col1a2** **Vim** **Col1a1** **Aebp1** **Timp2** **Selm** **Emp3** Prrx1	**Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** **Emp3** **Aebp1** Xist **Timp2** **Selm**
Trophoblast	**Cks1b** **Hmgb2** **Hsp90aa1** **Hmgn5** **Sri** **Cenpa** **Dbi** Mbnl3 Krt8 Phlda2	**Dbi** **Sri** **Hsp90aa1** S100a10 **Cks1b** **Hmgb2** **Hmgn5** Ybx3 Cyb5r3 **Cenpa**
iPSC	**Dppa5a** **Hsp90aa1** **Mybl2** **Nanog** **Ooep** **Sox2** **Zfp42** **Ckb** **Cdc5l** Tcl1	**Dppa5a** **Hsp90aa1** **Mybl2** **Ooep** **Ckb** **Nanog** **Cdc5l** **Zfp42** **Sox2** Fmr1nb

Genes of the model (100)

Data/data_train.h5ad, alphabetical order, coloured by selection role. Italics: no edge inside the selection (probability ≥ selection_edge_prob, literature-feasible; 0 genes). Trajectory preservation (classical OT on these genes vs every gene; network otvelo_granger): velocity cosine 0.88 (random genes 0.77), fate JS distance …

Aebp1	Dppa5a	Krt7	Rrm1
Antxr1	Efemp2	Krt8	Rrp9
Batf3	Ehd2	Laptm4a	S100a10
Bbx	Elf3	Ltbr	S100a14
Bhlhe40	Emp3	Marcksl1	Sat1
Btg2	Epcam	Mbnl3	Selm
Ccl2	Eva1b	Mgp	Sgol1
Ccnb1	Fam20c	Msmo1	Sms
Ccnd2	Fbn1	Mybl2	Sox2
Cd44	Flna	Myc	Sparc
Cd81	Fmr1nb	Nanog	Sri
Cdc5l	Foxo4	Nasp	Tagln
Cdkn1c	Gata2	Nfic	Tcl1
Cenpa	Gnl3	Nfix	Tgif1
Cited2	Gpx3	Nppb	Thbs1
Ckb	Hcfc1r1	Obox6	Timp2
Cks1b	Hmgb2	Ooep	Timp3
Col1a1	Hmgn5	Pcolce2	Tmbim1
Col1a2	Hsp90aa1	Phlda2	Tnc
Crabp2	Id3	Prnp	Tpm1
Cyb5r3	Ier3	Prrx1	Tspo
Dbi	Igfbp3	Pttg1ip	Vim
Ddit3	Kctd12	Rhoc	Xist
Ddx21	Klf4	Rhox6	Ybx3
Dmrtc2	Krt18	Rhox9	Zfp42

perturbed (2) query (27) driver (32) entropy (39)

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

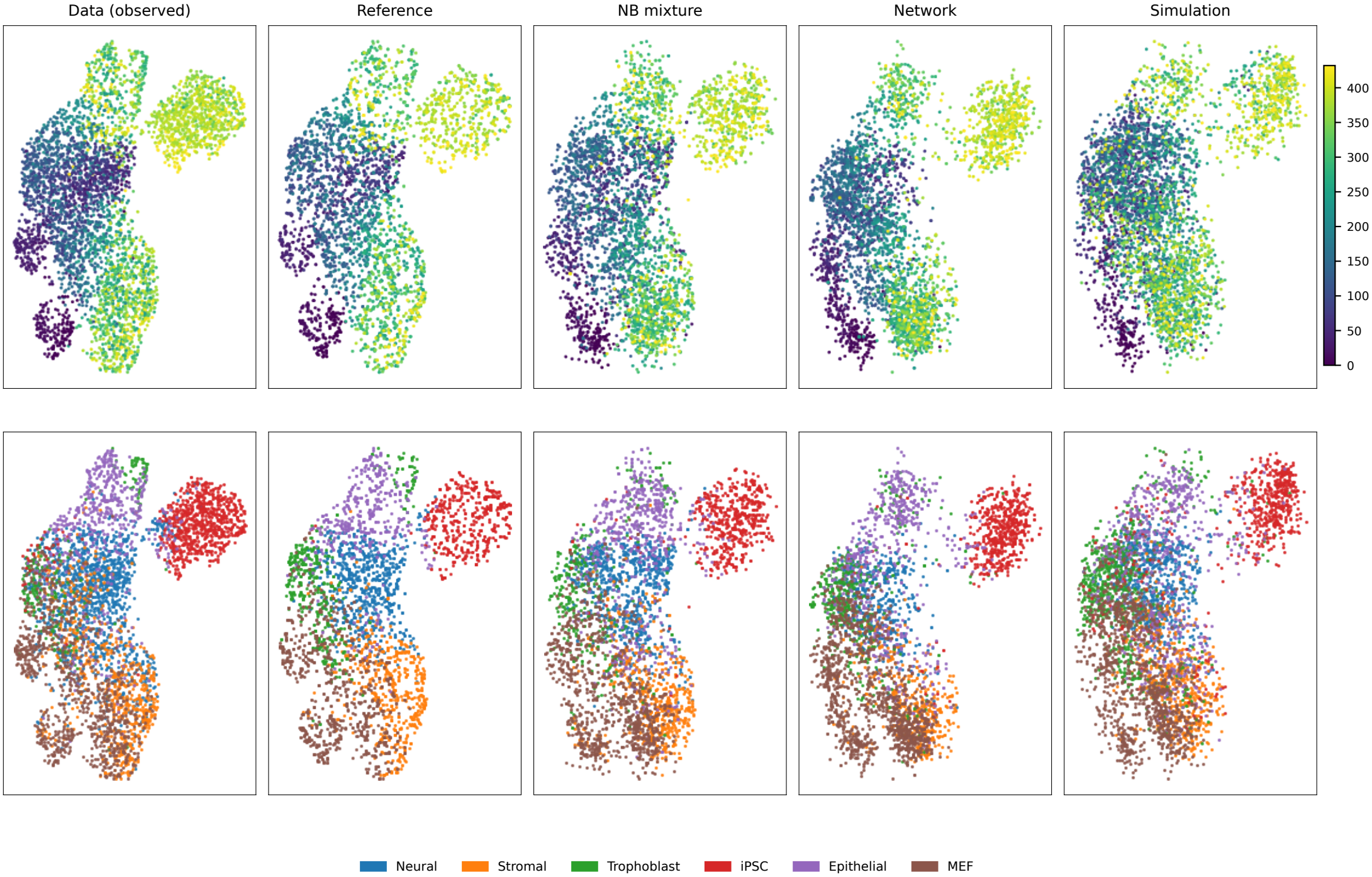
not estimated

Reason

4329 genes < 10000

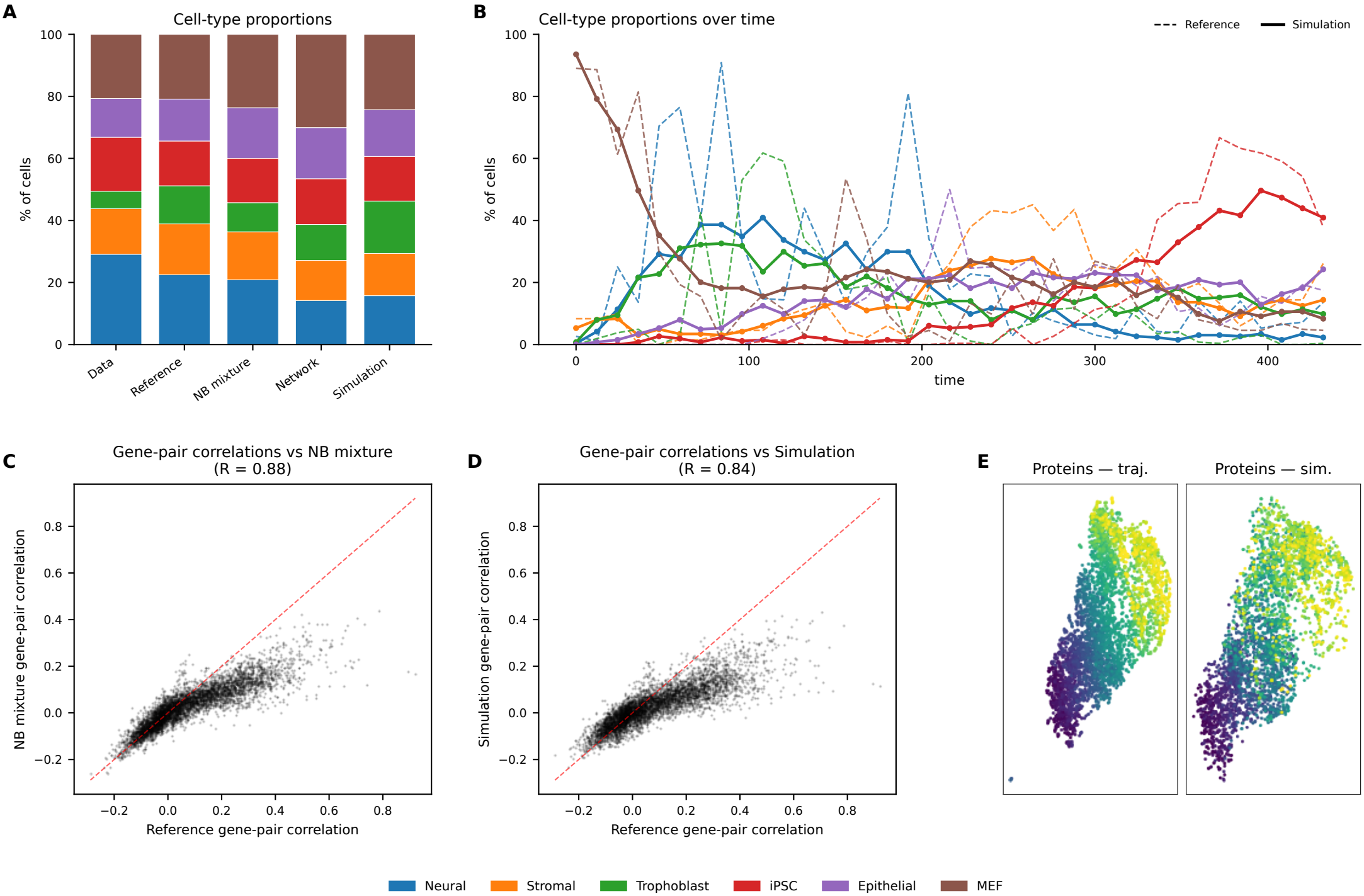
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. growth-weighted by exp ∫R_opt, as the simulation has proliferation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



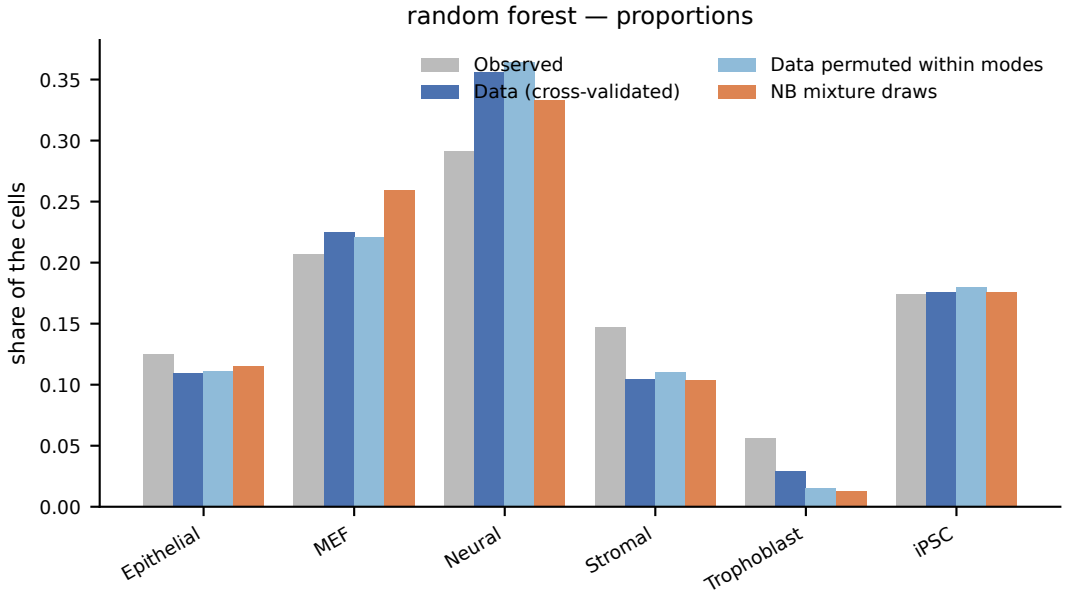
1. Generative model — quantitative checks

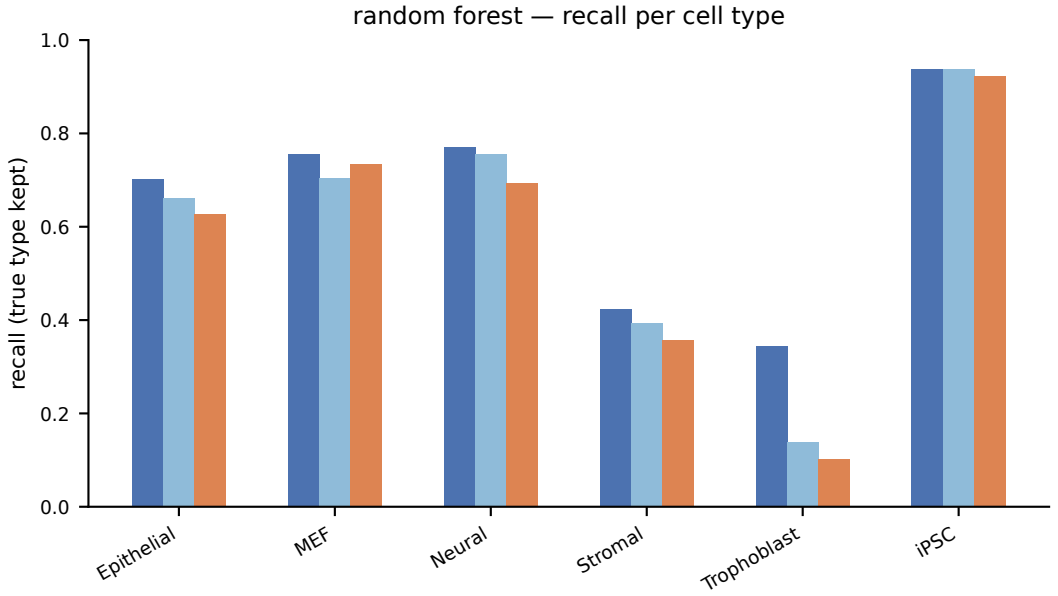
Reference = RNA of the trajectories equivalent to the simulation (growth-weighted: simulation with proliferation).

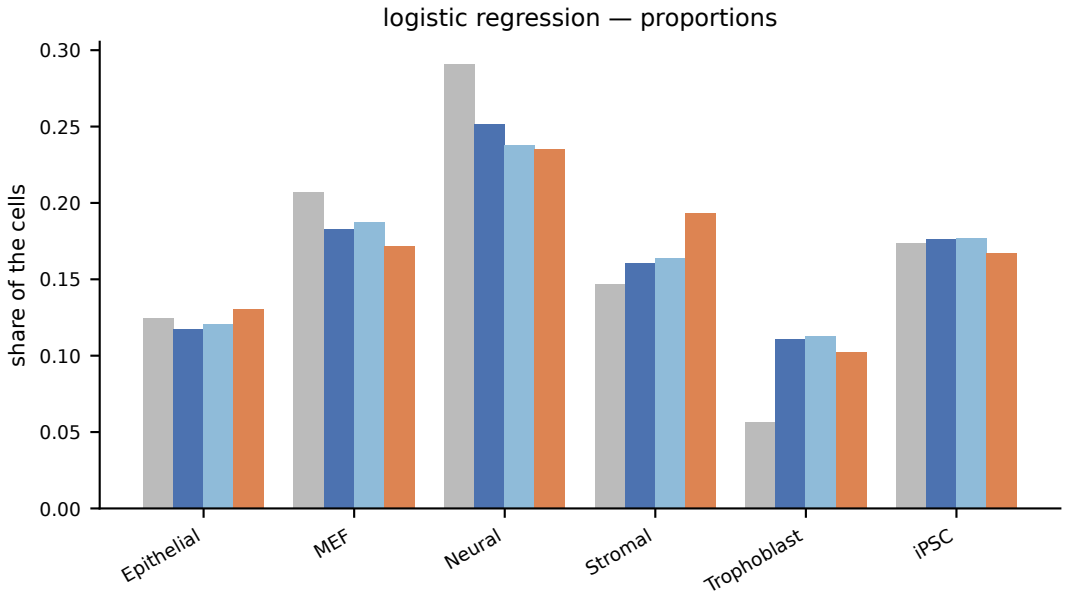


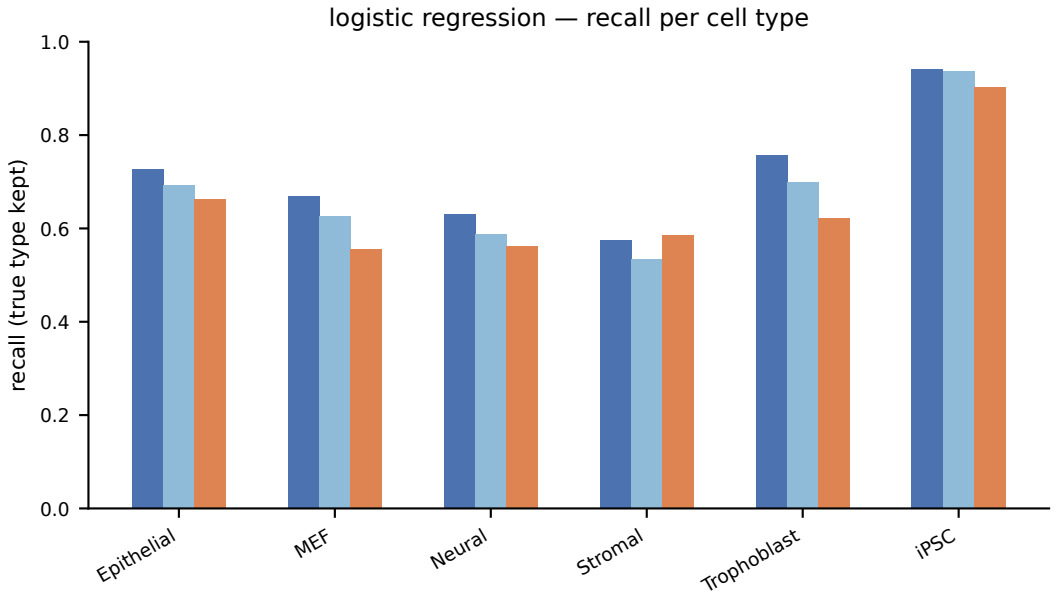
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.





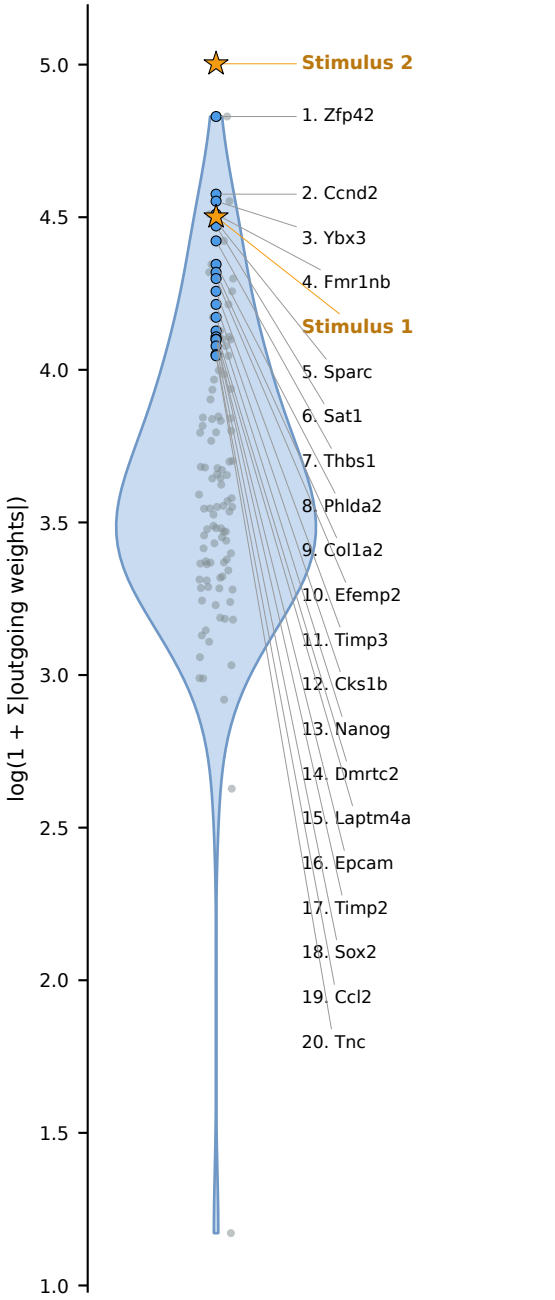




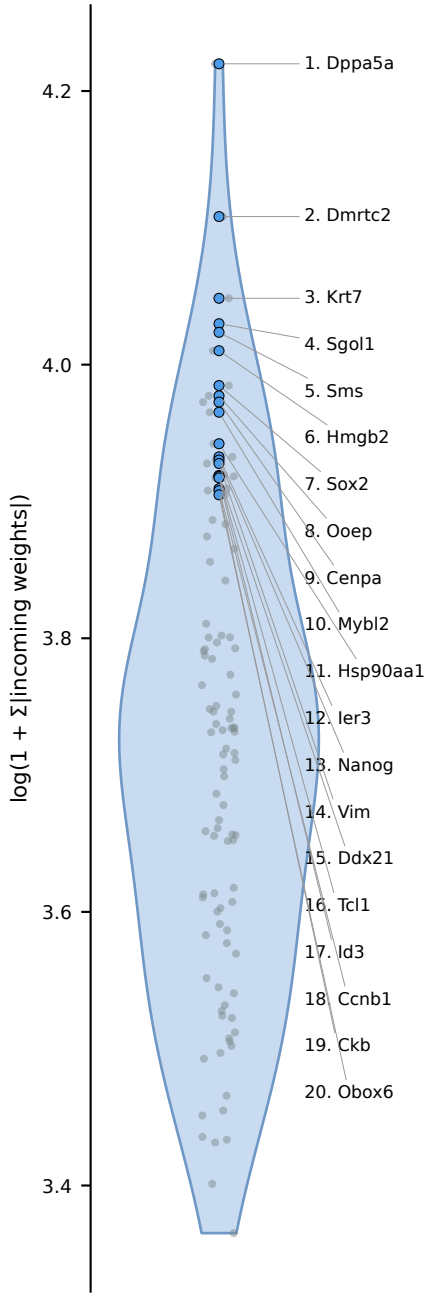
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

A



B



C Top 20 regulators

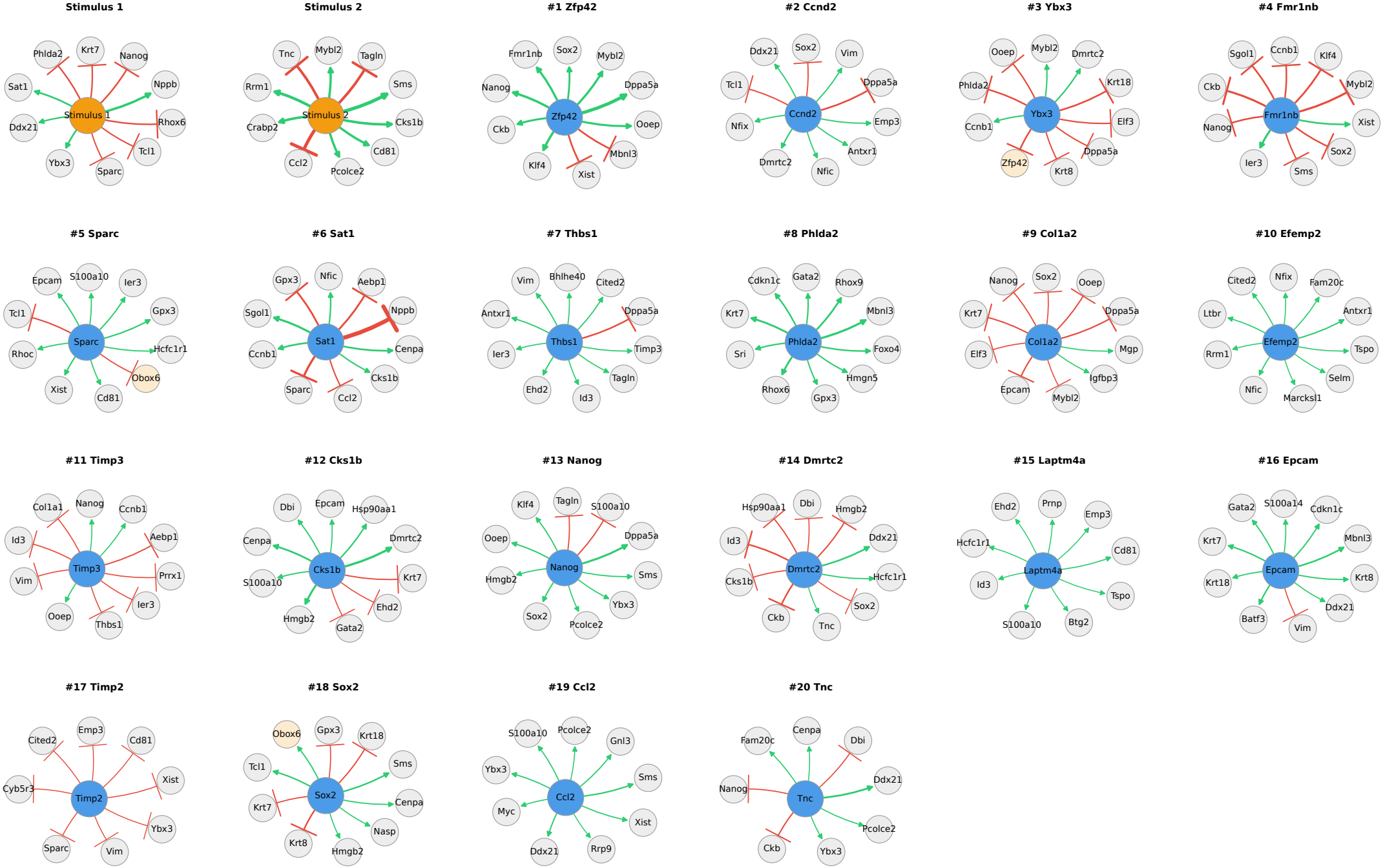
#	Regulator	Σ|w|	act.	inh.
1	Zfp42	124.20	36	60
2	Ccnd2	96.08	77	22
3	Ybx3	93.88	36	63
4	Fmr1nb	89.92	53	43
5	Sparc	87.73	69	29
6	Sat1	86.48	26	68
7	Thbs1	82.29	72	27
8	Phlda2	76.13	59	37
9	Col1a2	74.16	46	53
10	Efemp2	72.63	87	12
11	Timp3	69.62	42	56
12	Cks1b	66.65	52	47
13	Nanog	63.85	39	60
14	Dmrtc2	61.01	39	55
15	Laptm4a	59.83	70	29
16	Epcam	59.35	37	62
17	Timp2	59.29	21	78
18	Sox2	58.02	51	46
19	Ccl2	56.39	72	27
20	Tnc	56.17	60	37
★	Stimulus 1	89.12	35	65
★	Stimulus 2	147.81	49	36

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Dppa5a	67.03	51	42
2	Dmrtc2	59.84	44	47
3	Krt7	56.31	43	54
4	Sgol1	55.26	49	48
5	Sms	54.90	45	51
6	Hmgb2	54.16	47	51
7	Sox2	52.77	46	51
8	Ooep	52.37	55	39
9	Cenpa	52.12	44	54
10	Mybl2	51.73	52	46
11	Hsp90aa1	50.53	54	43
12	Ier3	50.04	42	56
13	Nanog	49.92	42	53
14	Vim	49.79	43	54
15	Ddx21	49.35	43	42
16	Tcl1	49.32	45	51
17	Id3	49.26	49	49
18	Ccnb1	48.87	43	52
19	Ckb	48.80	44	55
20	Obox6	48.64	44	48

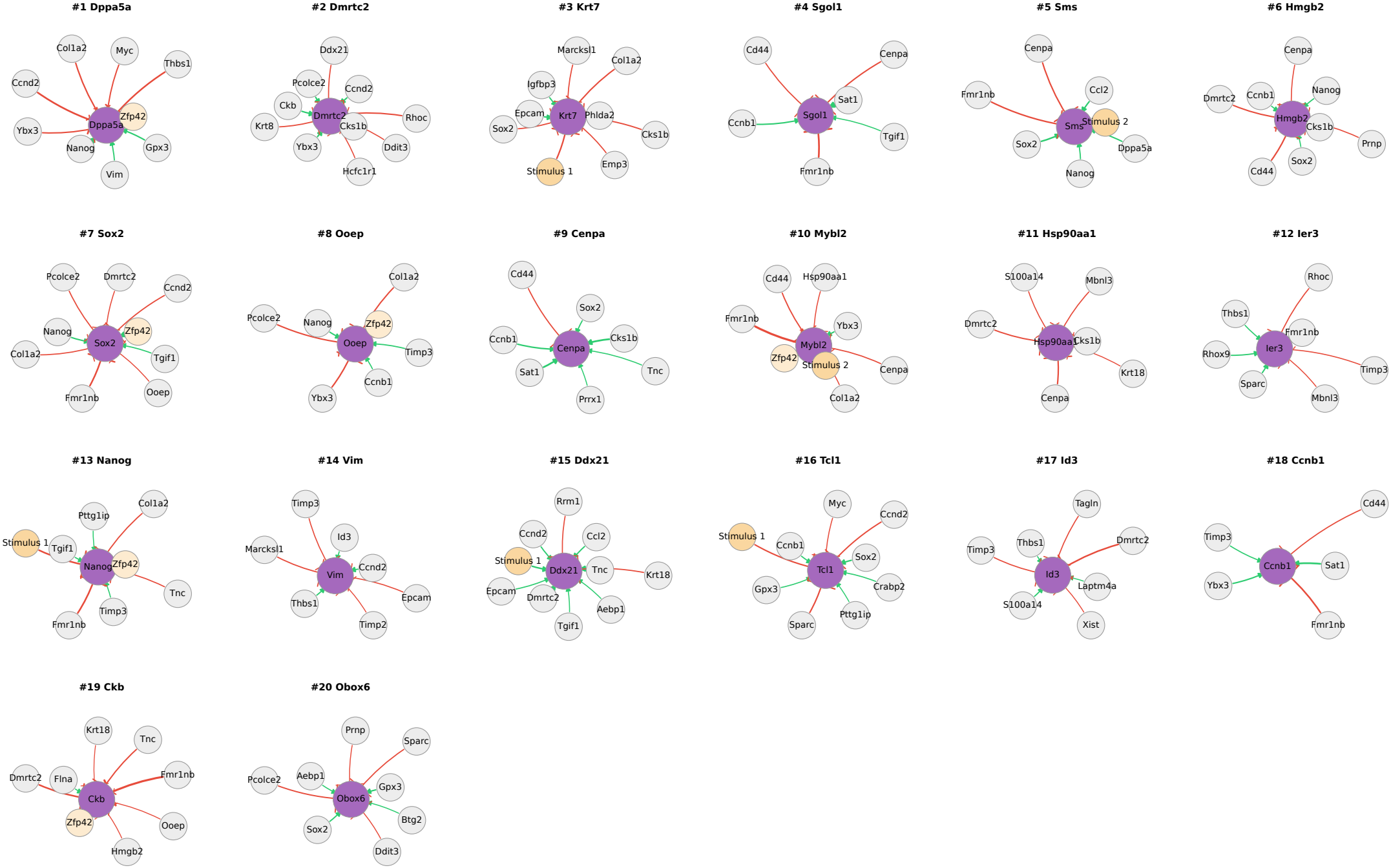
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



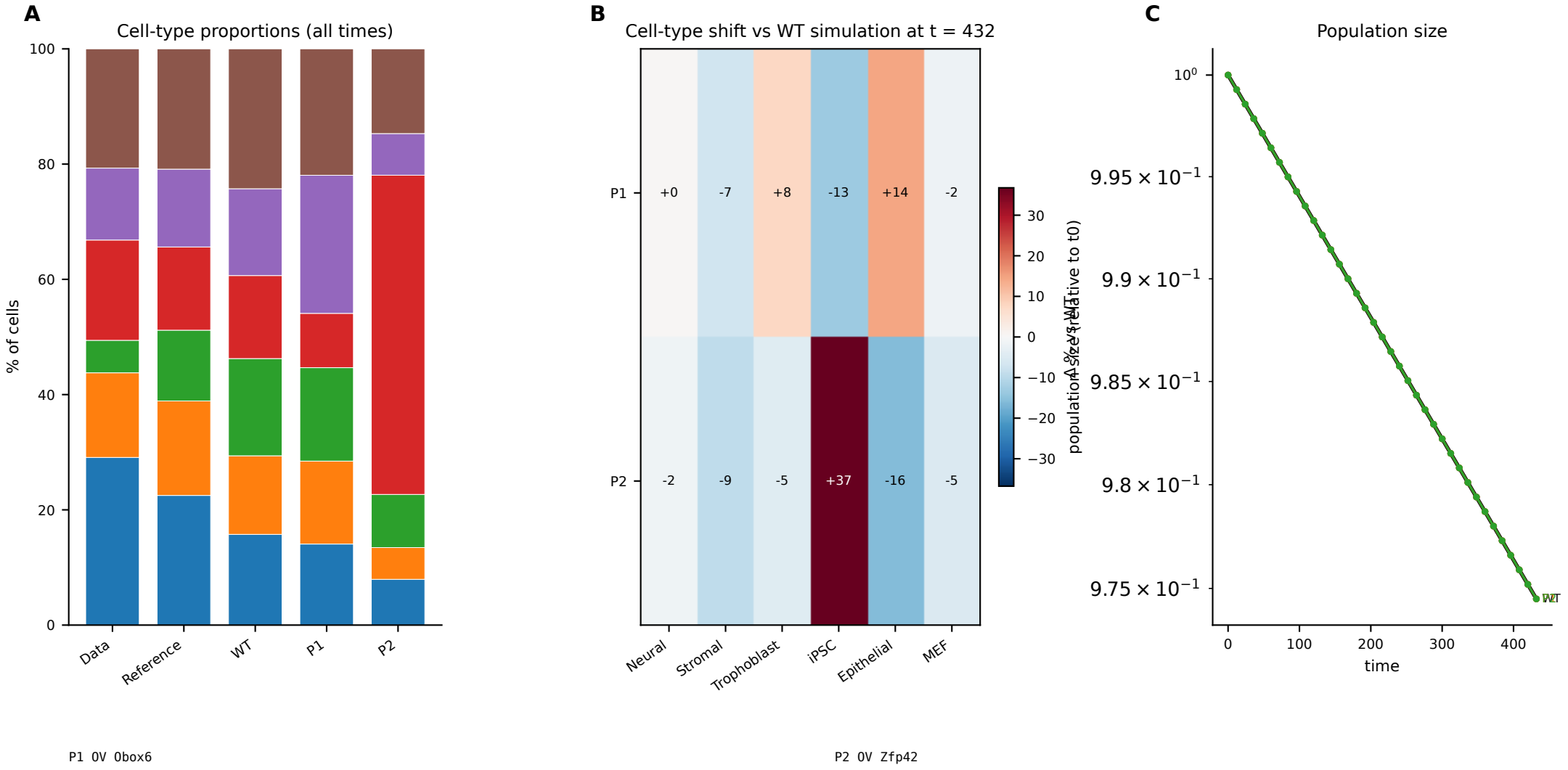
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

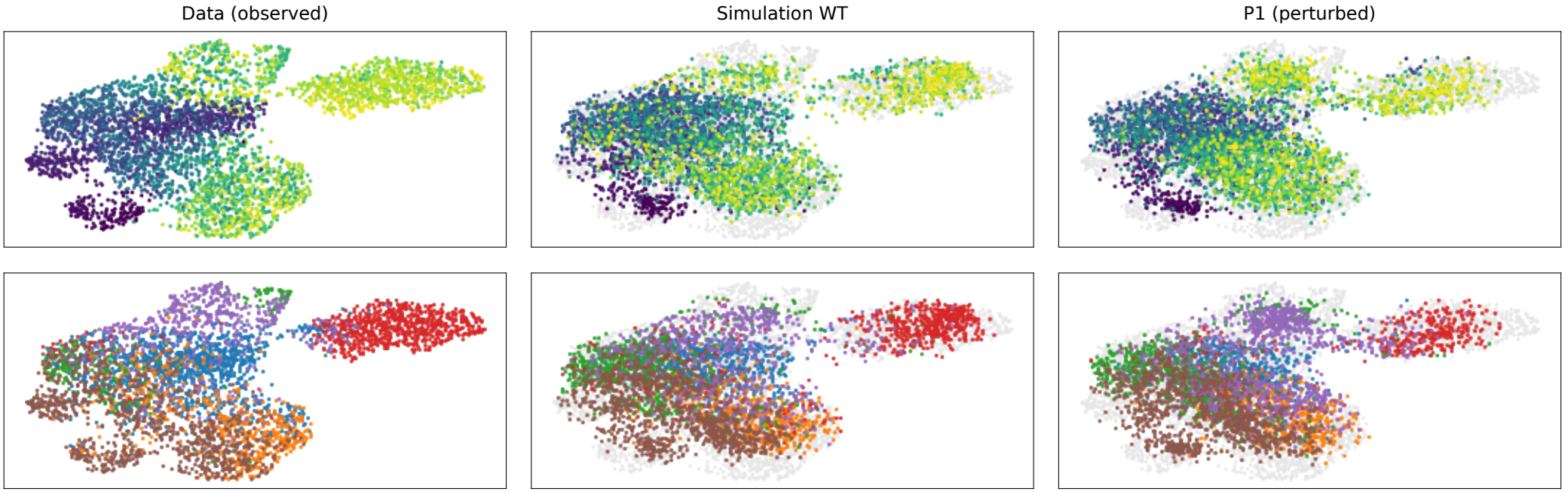


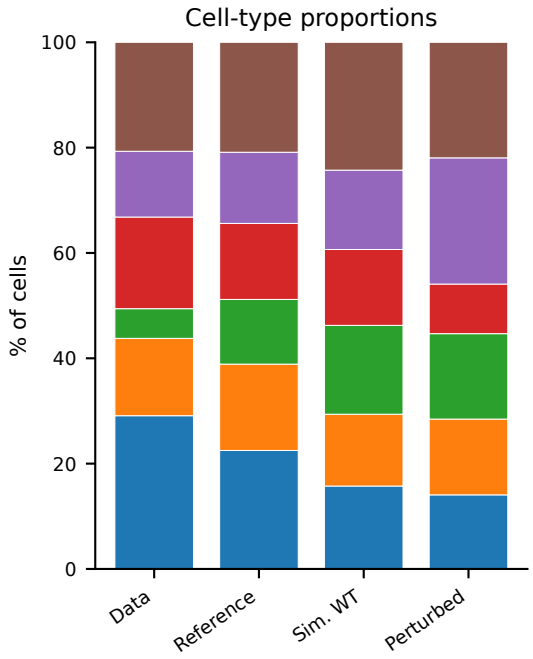
P1 OV Obox6

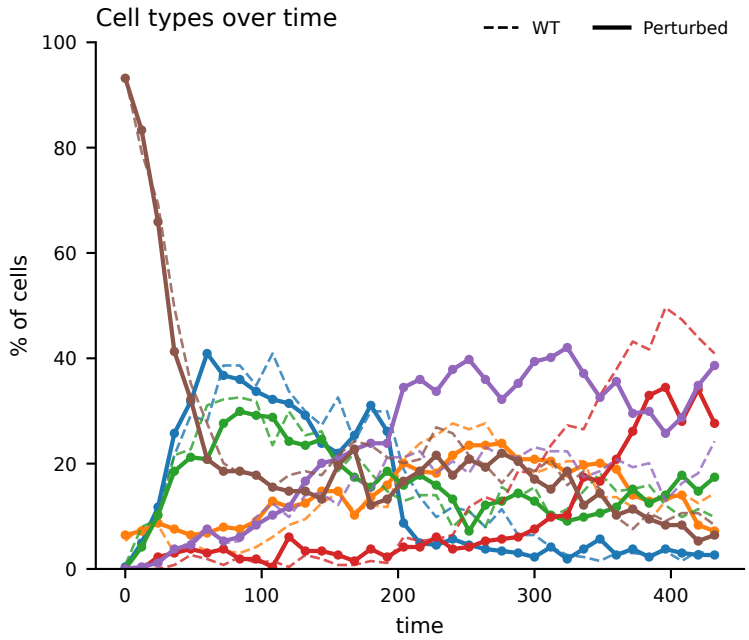
P2 OV Zfp42

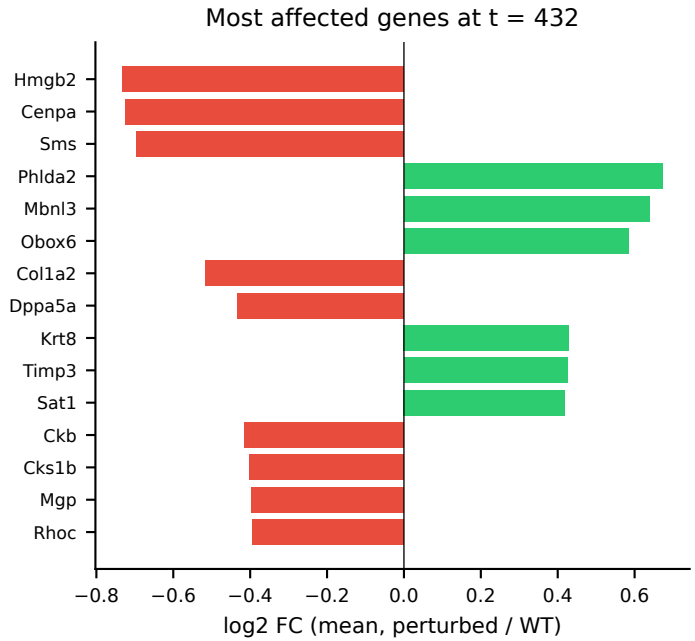
3. Perturbation P1 — OV Obox6

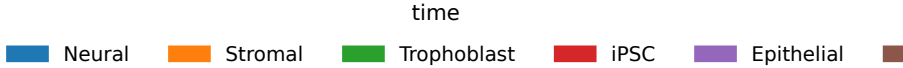
KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 432: ×1.00 vs WT.





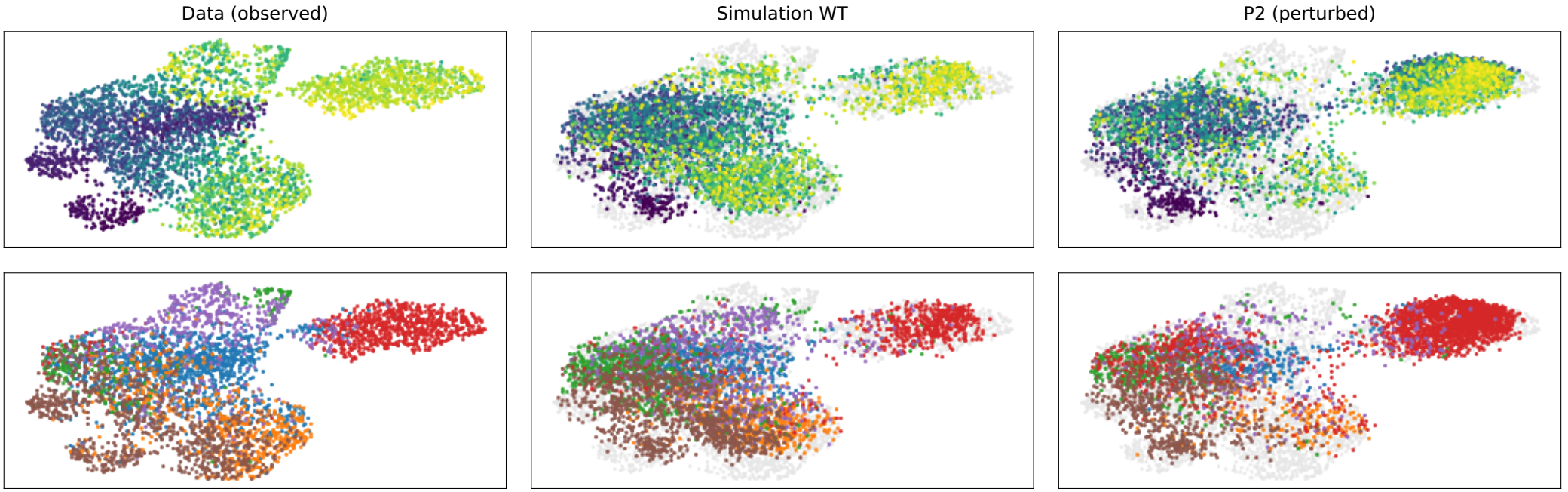


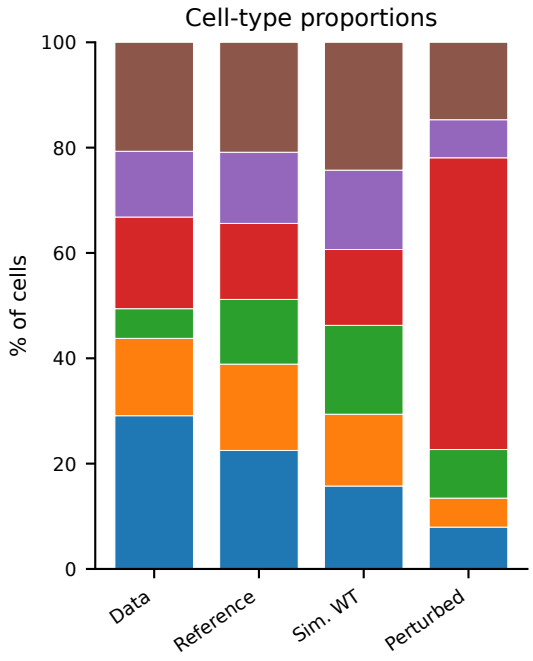


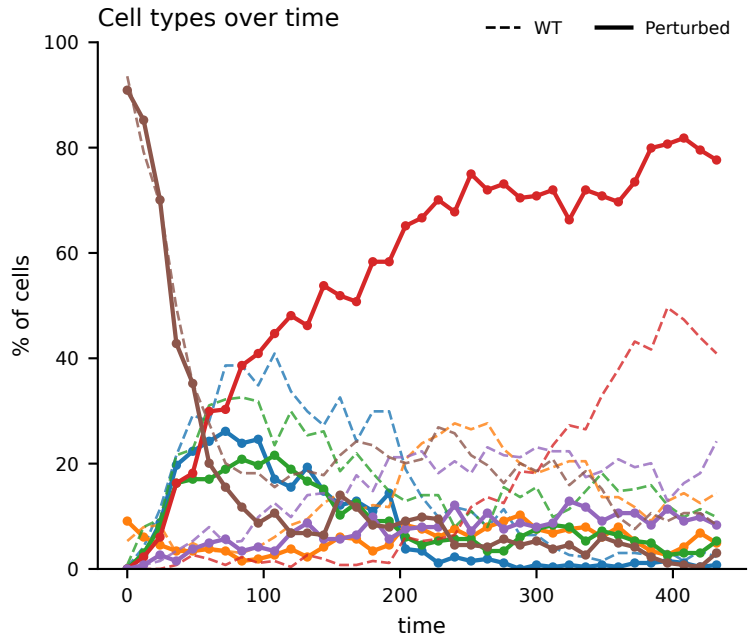


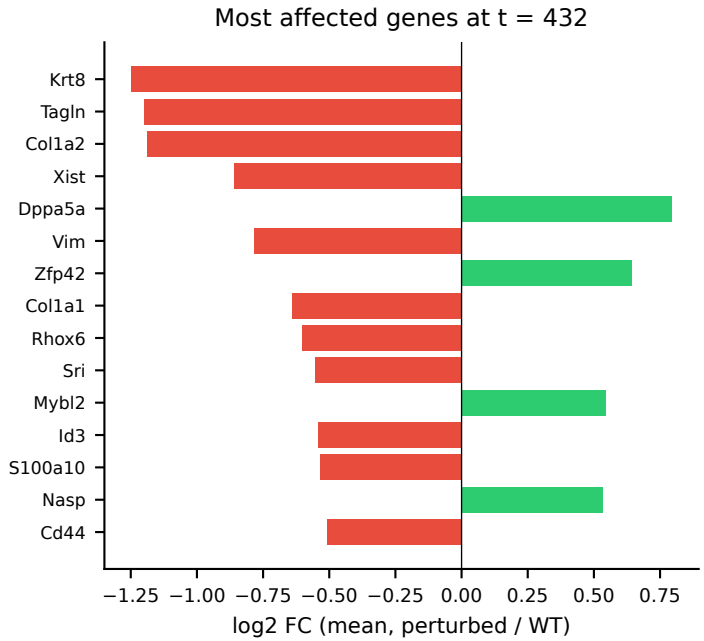
3. Perturbation P2 — OV Zfp42

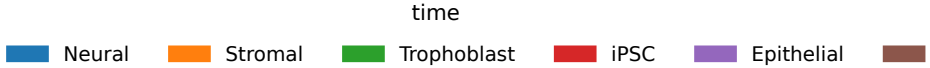
KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 432: ×1.00 vs WT.







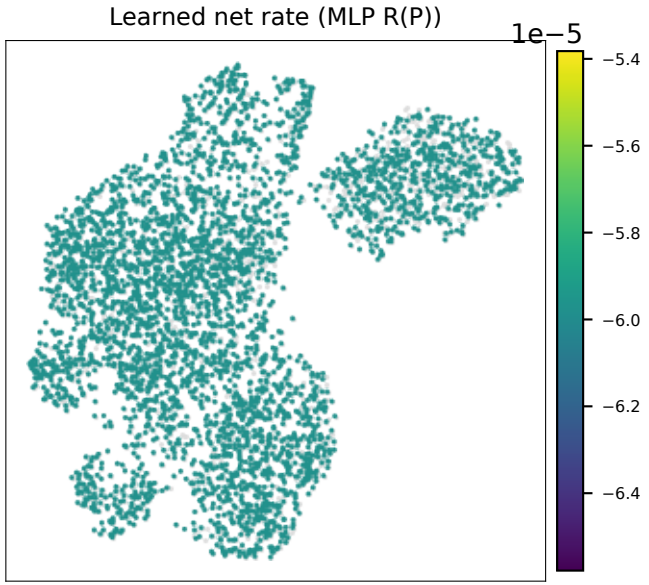


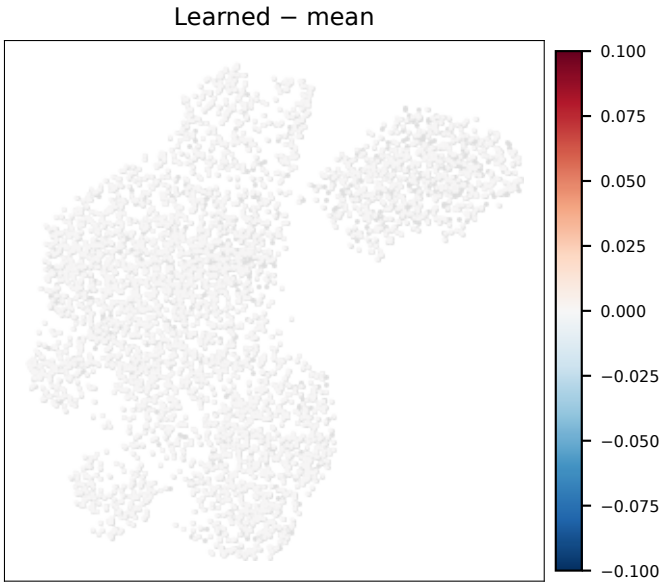


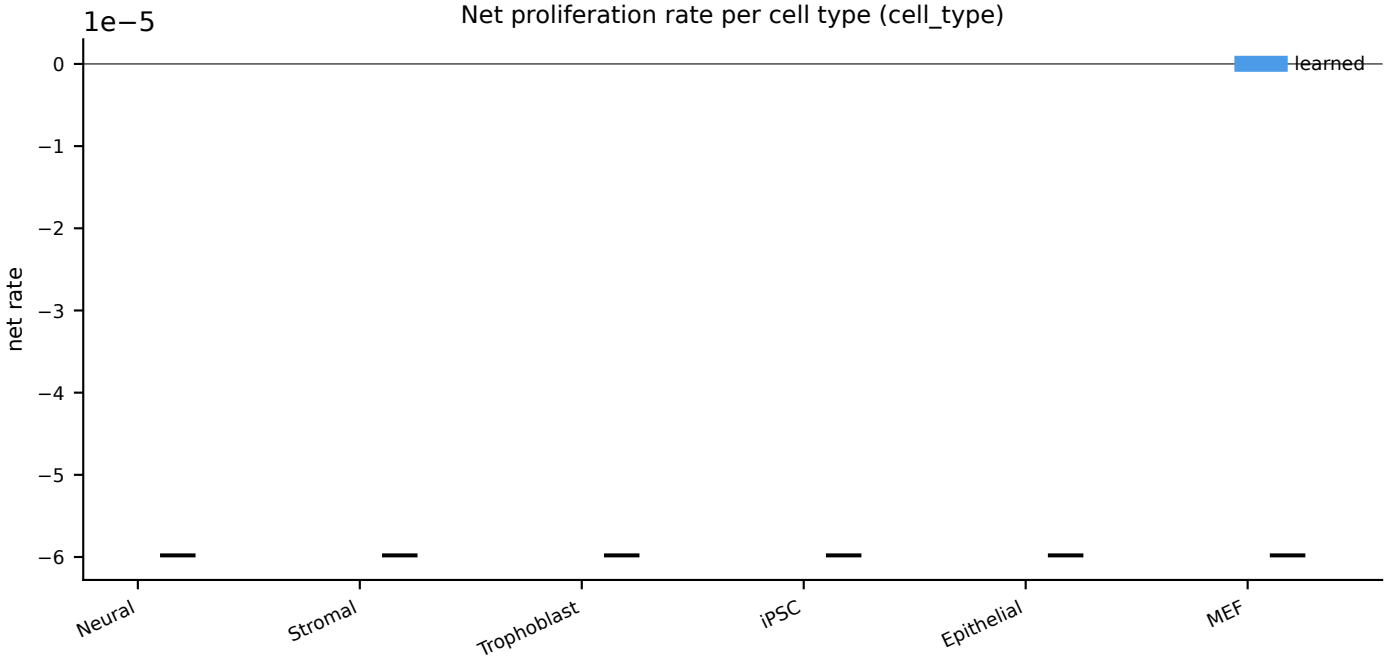
4. Proliferation — prior vs learned net rate (birth − death)

Learned = MLP R(P), mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)





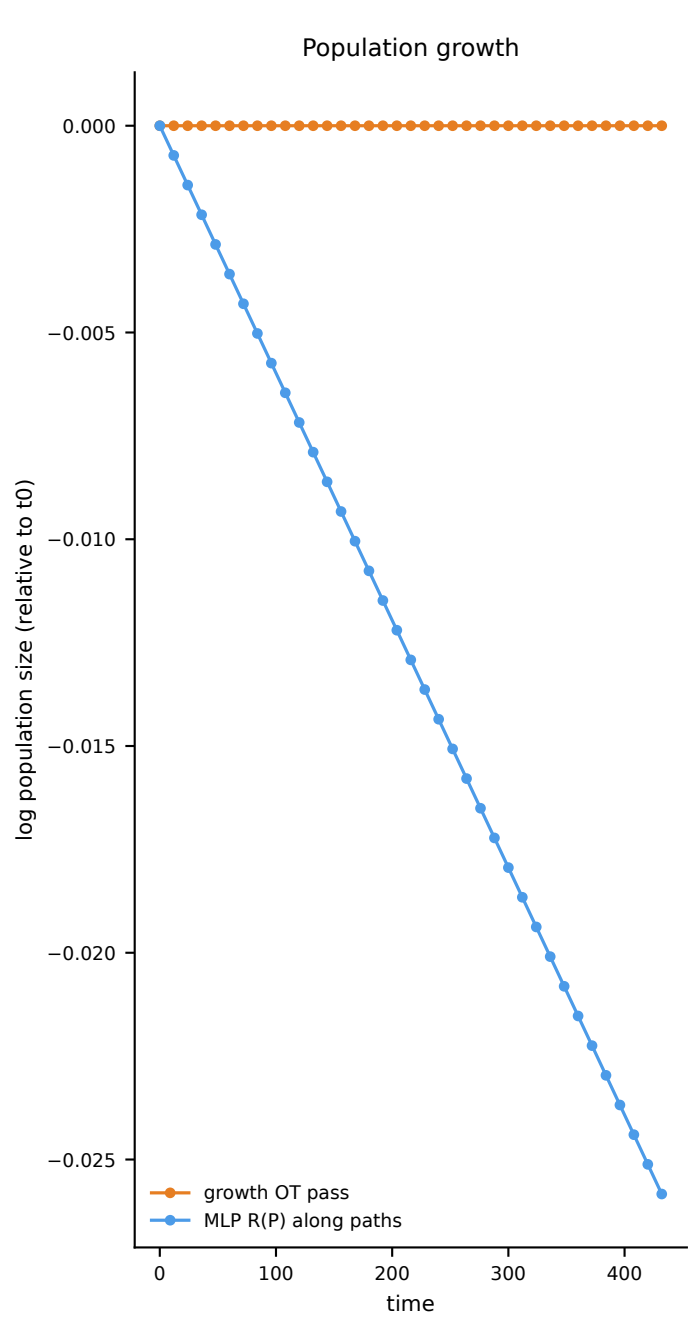


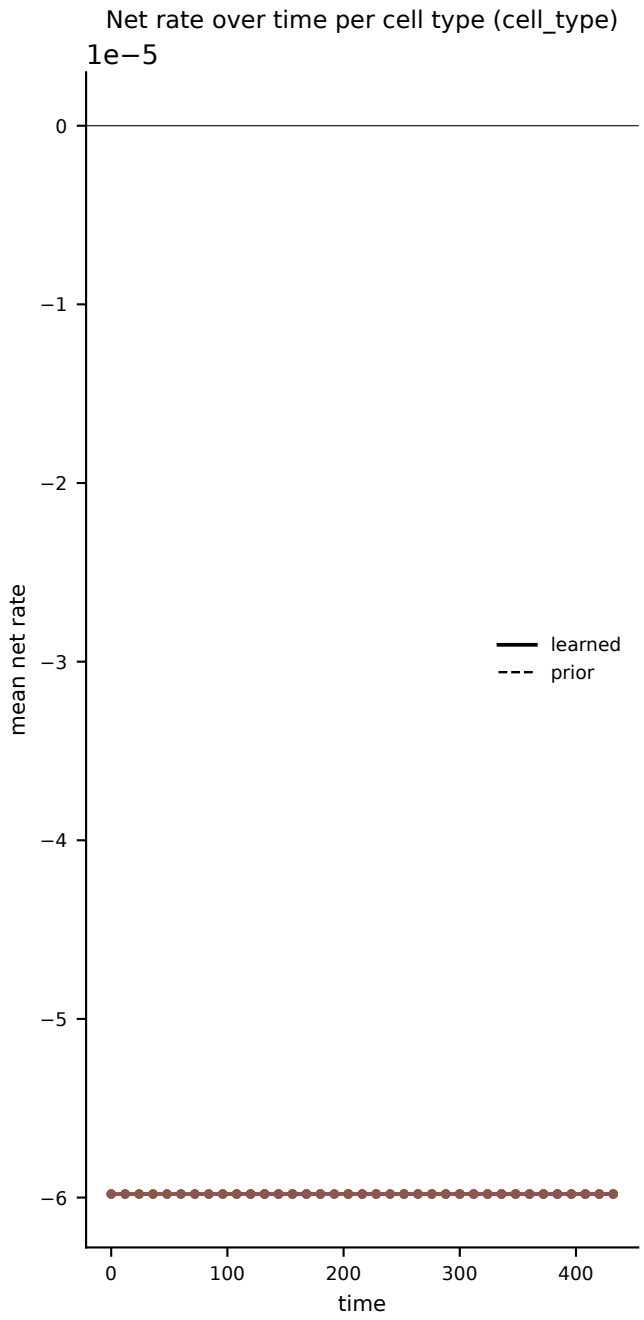
Mean net rate per cell type (cell_type)

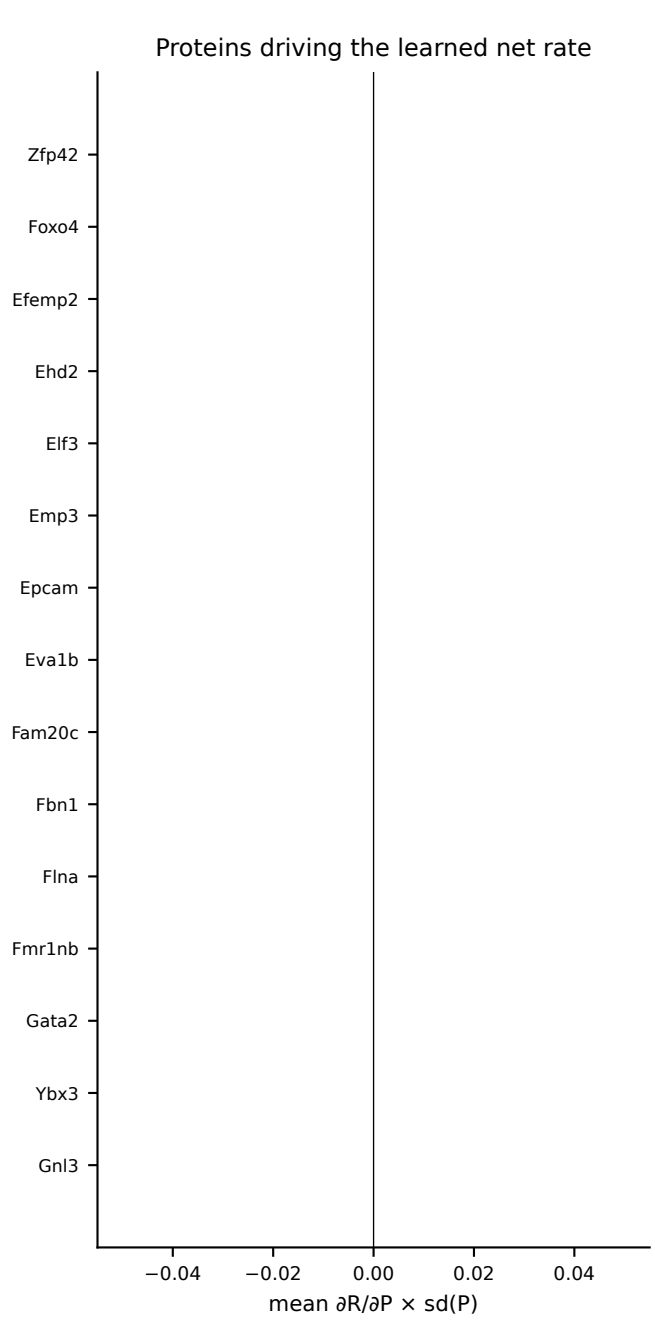
Cell type	Anchor	Prior	Learned
Neural	—	—	-5.98e-05
Stromal	—	—	-5.98e-05
Trophoblast	—	—	-5.98e-05
iPSC	—	—	-5.98e-05
Epithelial	—	—	-5.98e-05
MEF	—	—	-5.98e-05

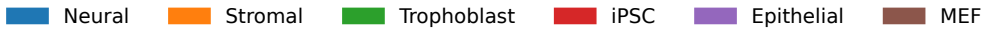
4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).



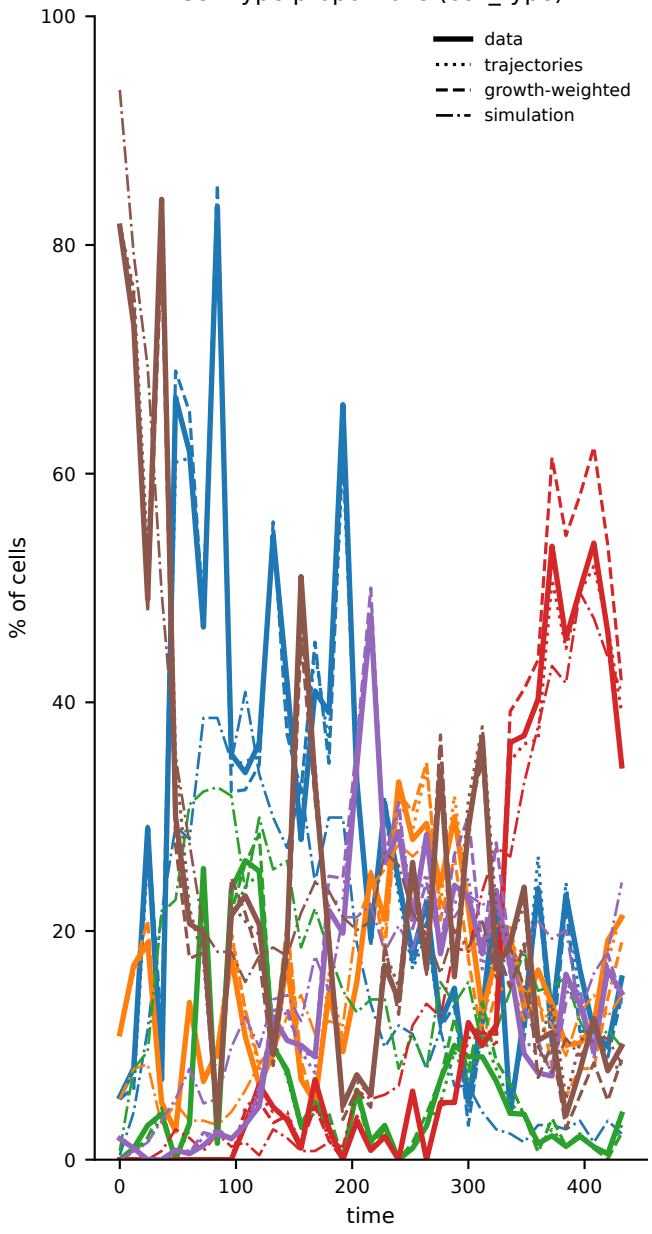


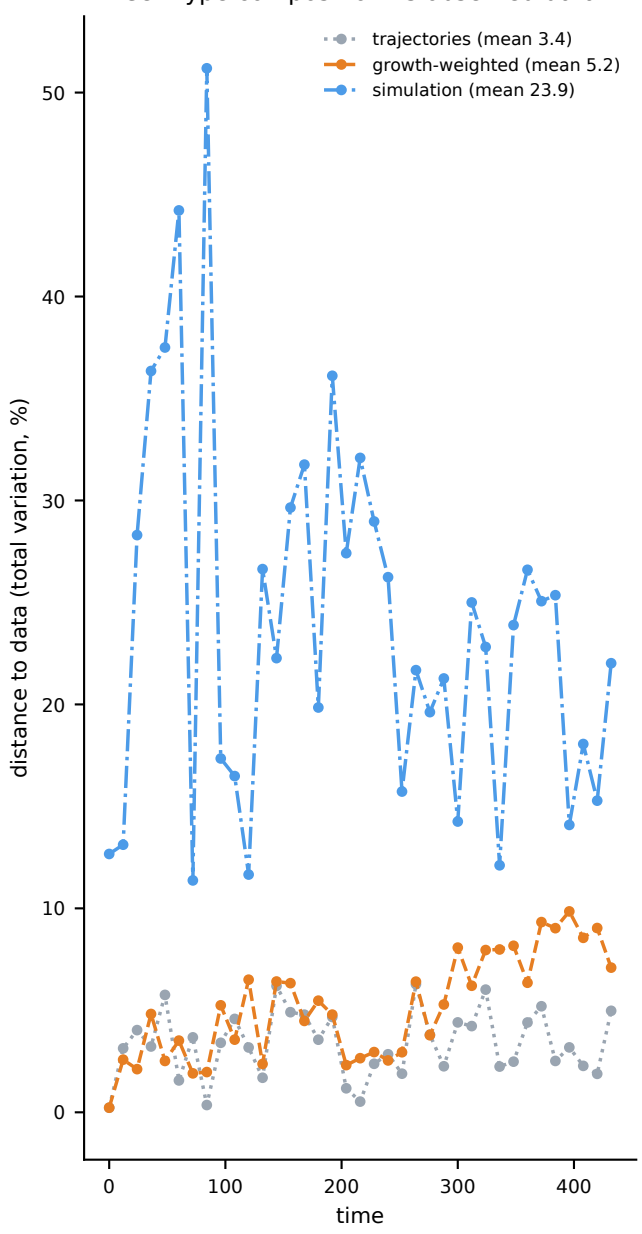


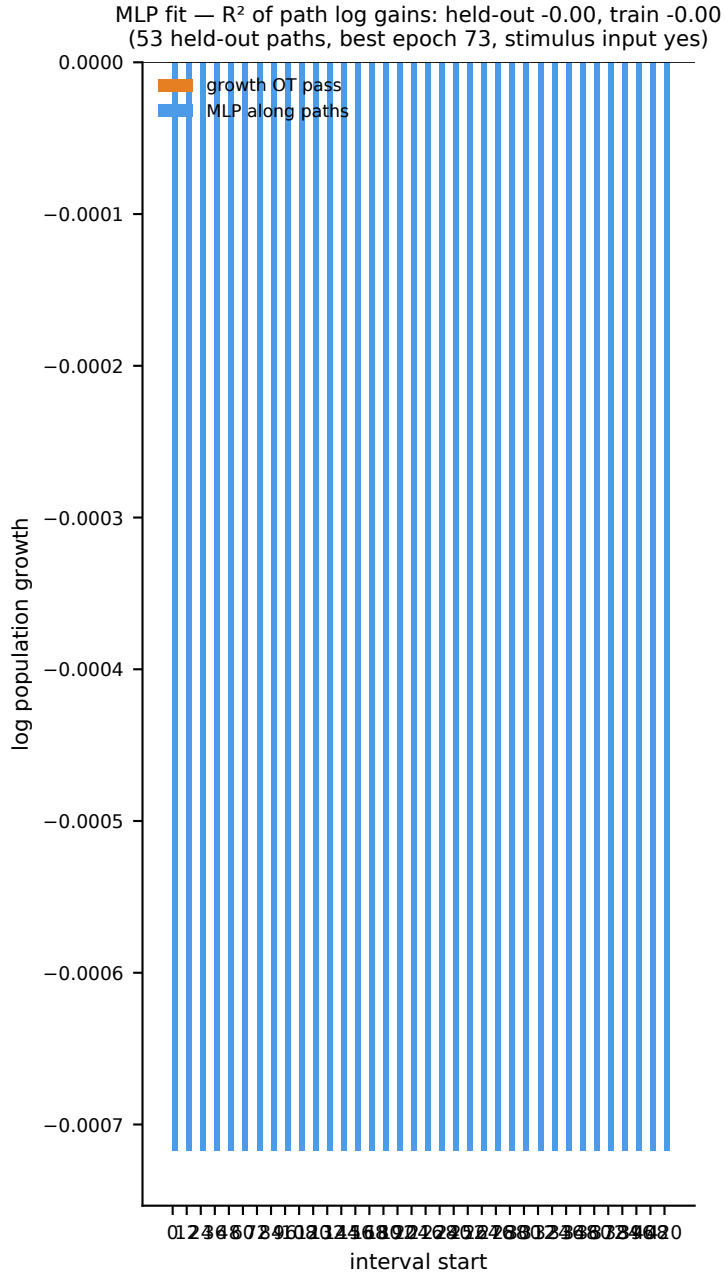


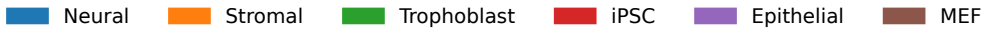
4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data. The simulation used the proliferation MLP.









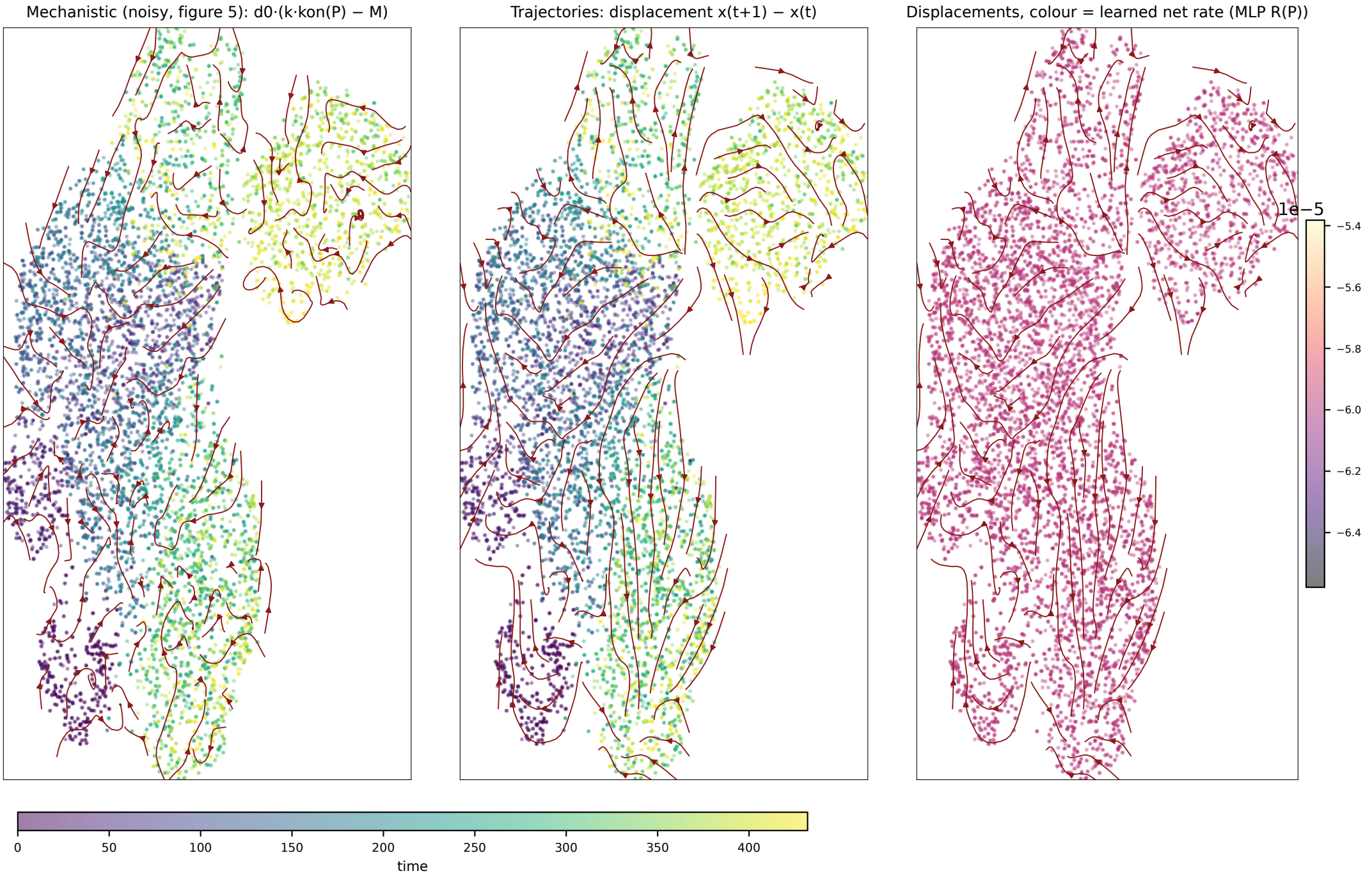
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

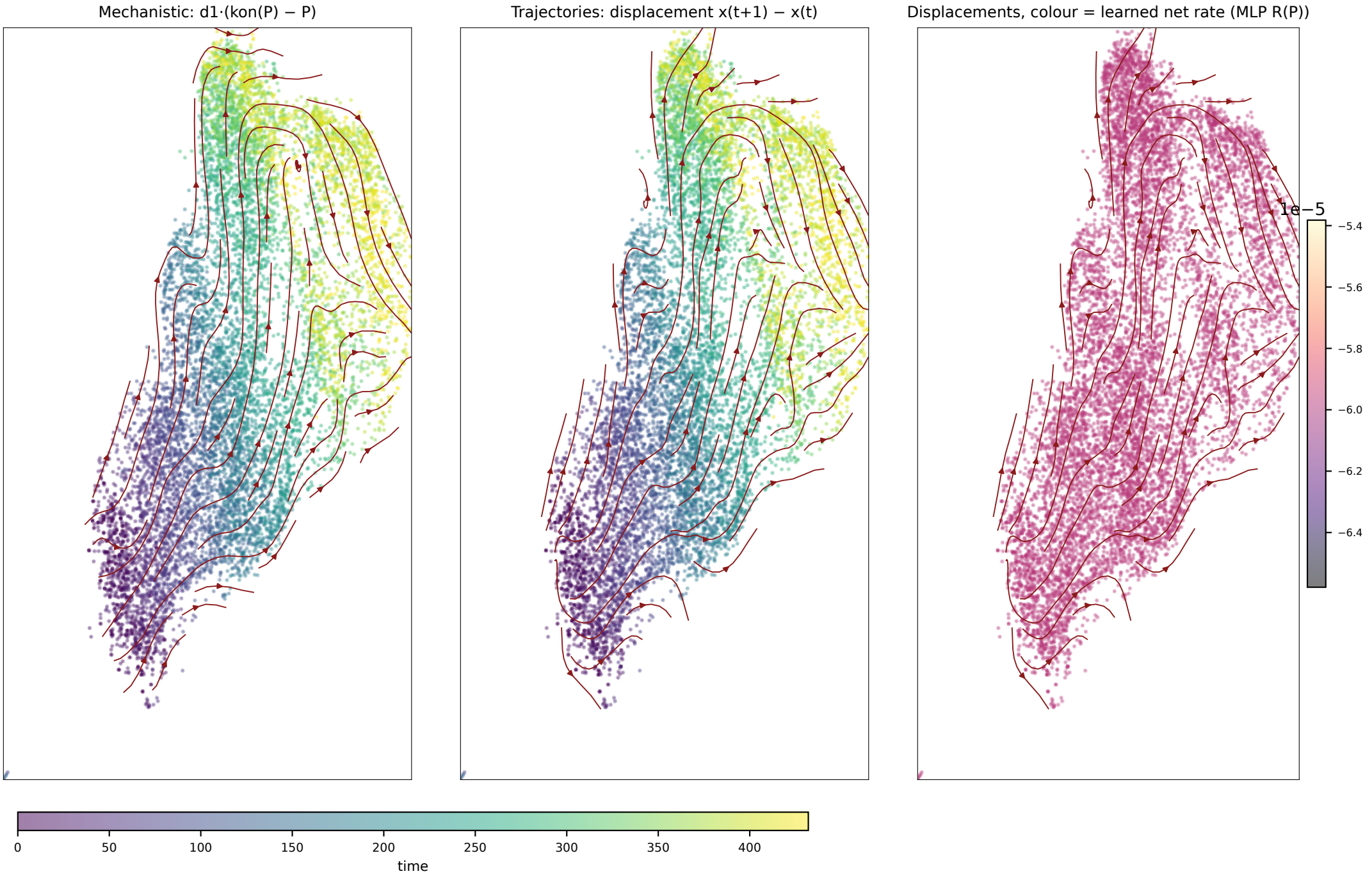
5. Learned dynamics — mRNA velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.09.



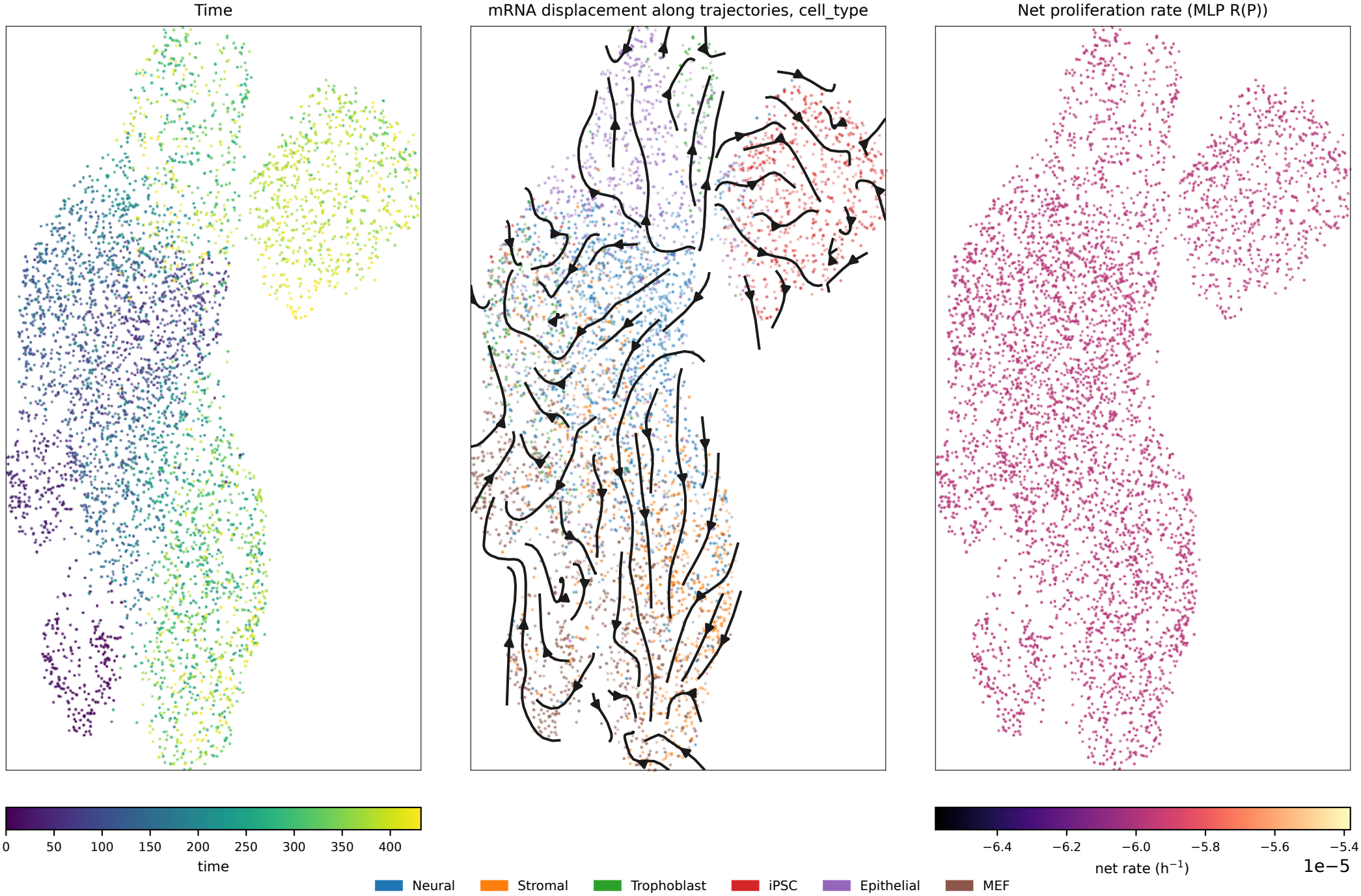
5. Learned dynamics — Proteins velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.52.



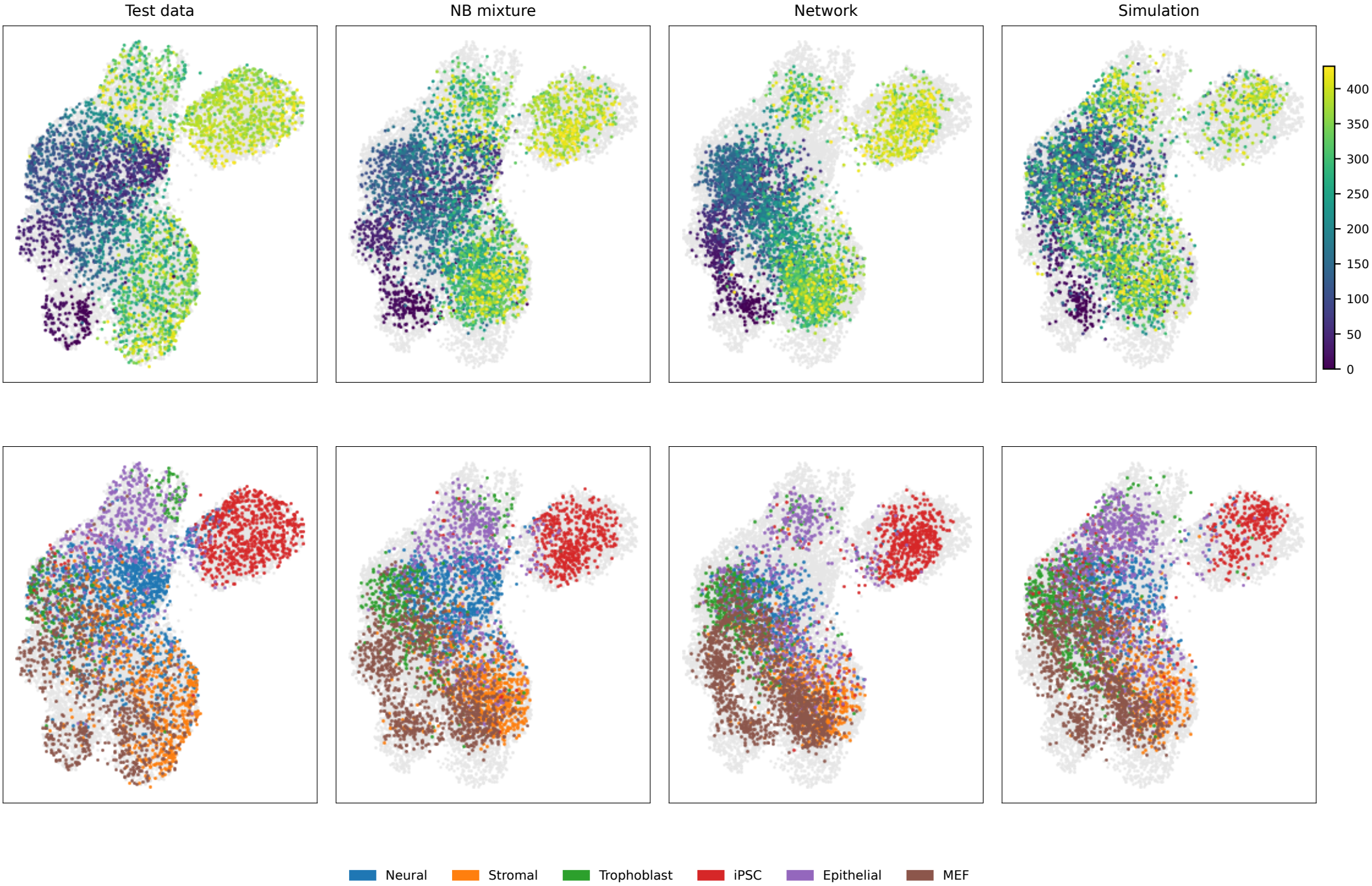
5. Learned dynamics — summary

All 9768 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

